

ON THE CARDINALITY OF MEASURES OF MAXIMAL RELATIVE ENTROPY FOR SMOOTH SKEW PRODUCTS

MATHEUS M. CASTRO* AND GARY FROYLAND*

ABSTRACT. Let Ω and M be compact smooth manifolds and let $\Theta : \Omega \times M \rightarrow \Omega \times M$ be a $\mathcal{C}^{1+\alpha}$ skew-product diffeomorphism over an Anosov base. We show that Θ has at most countably many ergodic hyperbolic measures of maximal relative entropy. When $\dim M = 2$, if Θ has positive relative topological entropy, then Θ has at most countably many ergodic measures of maximal relative entropy.

CONTENTS

1. Introduction and main results	2
Structure of the paper.	3
1.1. Measures of maximal relative entropy	3
1.2. Main results	5
1.3. An application of Corollary B	6
2. Hyperbolic measures and Borel homoclinic classes	7
2.1. Hyperbolic measures	7
2.2. Borel homoclinic classes	9
3. Topological Markov shifts	9
4. Measures of maximal relative entropy on topological Markov shifts	19
4.1. Petersen-Quas-Shin construction	21
5. Proofs of Theorem A, Corollary B and Proposition 1.6	27
Acknowledgements	32
References	32

*SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW SOUTH WALES, SYDNEY, NSW 2052, AUSTRALIA

E-mail addresses: m.manzatto-de-castro@unsw.edu.au, g.froyland@unsw.edu.au.

Date: February 15, 2026.

2020 Mathematics Subject Classification. 28D20; 37D25; 37D35; 37H05.

Key words and phrases. Measures of maximal relative entropy; skew products; random dynamical systems; hyperbolic dynamical systems; thermodynamic formalism.

1. INTRODUCTION AND MAIN RESULTS

For dynamical systems related by a factor map, the natural analogue of maximal entropy is the concept of maximal relative entropy. Let (X, f) and (Y, g) be continuous maps on compact metric spaces and let $\pi: X \rightarrow Y$ satisfy $g \circ \pi = \pi \circ f$. Write $\mathcal{M}(f)$ and $\mathcal{M}(g)$ for the sets of invariant Borel probability measures of f and g , respectively. Given $\nu \in \mathcal{M}(g)$, a measure $\mu \in \mathcal{M}(f)$ is a measure of maximal ν -relative entropy if $\pi_*\mu = \nu$ and

$$h_\mu(f) = \sup\{h_\eta(f) : \eta \in \mathcal{M}(f), \pi_*\eta = \nu\}.$$

Equivalently, if $h_\nu(g) < \infty$ the Abramov–Rokhlin formula states that μ maximises the conditional entropy

$$h_\eta(f \mid \nu) := h_\eta(f \mid \pi^{-1}\mathcal{B}_Y) = h_\eta(f) - h_\nu(g)$$

among all π -lifts η of ν . This relative viewpoint is captured by the relativised variational principle of Ledrappier–Walters and in Walters’ compensation function, extending the notion of relative entropy to relative equilibrium states [33, 58].

The general picture of measures of maximal relative entropy and relative equilibrium states is well advanced in symbolic dynamics, particularly for factor maps between shifts where X is an irreducible subshift of finite type. Petersen–Quas–Shin proved that for each ergodic base measure ν on Y there are only finitely many ergodic measures of maximal relative entropy on X , and they established explicit bounds on this number [48]. These bounds were improved in [2], introducing the notion of class degree. A subsequent work extended these advances to relative equilibrium states in [1]. A structure theorem for infinite-to-one factor codes $\pi: X \rightarrow Y$ between irreducible subshifts of finite type was proved in [59]: that there exists a sofic shift Y_1 , such that $\pi = \pi_2 \circ \pi_1$. This allowed Yoo to show that if ν is an equilibrium state on Y for a sufficiently regular potential, then there is a unique lift on X that is a relative equilibrium state [59].

Measures of maximal relative entropy and relative equilibrium states naturally arise in random dynamical systems. There, they are usually called quenched relative equilibrium states. Consider a probability-preserving base $(\Omega, \theta, \mathbb{P})$ and the skew-product map

$$\Theta : (\omega, x) \in \Omega \times M \mapsto (\theta(\omega), T_\omega(x)) \in \Omega \times M.$$

Observe that when taking $X = \Omega \times M$, $Y = \Omega$, $f = \Theta$, $g = \theta$, and $\pi(\omega, x) = \text{proj}_\Omega(\omega, x) = \omega$ we retrieve the setting of the first paragraph under the assumption that $h_\mathbb{P}(\theta) < \infty$.

Foundational results of Kifer established existence and uniqueness of relative equilibrium states for random uniformly expanding transformations and identified quenched relative pressure with the principal Lyapunov exponent of the random transfer-operator cocycle [31]. Bogenschütz developed parallel notions of metric entropy and pressure for random dynamical systems and proved a variational principle, in the framework commonly used nowadays [12]. Thermodynamic formalism for random systems has since been developed across a variety of settings: random symbolic dynamics [13, 25, 22, 54, 53]; smooth expanding maps [31, 42]; non-uniformly smooth expanding maps [5, 55]; and non-uniformly hyperbolic random interval maps (closed and open, with discontinuities) [8, 6, 7]; see also the references therein.

Although the quenched relative thermodynamic formalism theory for random dynamical systems is extensive, results for systems generated by diffeomorphisms remain scarce outside the SRB setting (for SRB results, see [32, 11, 24, 3, 39] and the references therein). In particular, except in settings where Gibbs-type bounds are available, there is no general method for bounding the number of measures of maximal relative entropy, let alone the number of relative equilibrium states, in these settings. In this paper, we study how many measures of maximal relative entropy a skew product $\Theta(\omega, x) = (\theta(\omega), T_\omega(x))$

over an Anosov base may admit. This question is part of an ongoing program to quantify the number of measures of maximal entropy across classes of systems.

In the setting of smooth diffeomorphisms, Newhouse proved that $\mathcal{C}^\infty$ diffeomorphisms on compact smooth manifolds always admit a measure of maximal entropy [43]. Buzzi later conjectured that for surface diffeomorphisms with positive topological entropy, there are at most countably many ergodic measures of maximal entropy [18, Conjecture 1]. Sarig’s groundbreaking symbolic coding for $\mathcal{C}^{1+\alpha}$ surface diffeomorphisms confirmed this countability and also settled a conjecture of Katok on periodic-orbit growth [50]. This result was recently sharpened by Buzzi–Crovisier–Sarig, proving that $\mathcal{C}^\infty$ surface diffeomorphisms with positive entropy have only finitely many ergodic measures of maximal entropy, and exactly one in the transitive case, [20], which answered another of Buzzi’s conjectures [18, Conjecture 2].

In higher dimensions, Ben Ovadia extended Sarig’s coding to non-uniformly hyperbolic diffeomorphisms on arbitrary manifolds, yielding countability results for hyperbolic measures of maximal entropy [10]. More recently, Buzzi–Crovisier–Sarig established the strong positive recurrence (SPR) property, and via SPR obtained strong statistical properties for the associated invariant measures, including exponential decay of correlations and an almost-sure invariance principle [21]. Related results are known for two-dimensional fibre skew products over hyperbolic bases; see Marín–Poletti–Veiga [41]. Analogous results are available for flows [38, 19, 36] and for maps with singularities and endomorphisms [35, 34, 4, 37].

Our contribution is a relative analogue of these countable cardinality results. For $\mathcal{C}^{1+\alpha}$ skew-product diffeomorphisms $\Theta(\omega, x) = (\theta(\omega), T_\omega(x))$ over a transitive Anosov base (Ω, θ) and for any ergodic θ -invariant measure $\mathbb{P}$ with full support, we prove that there are only countably many hyperbolic ergodic Θ -invariant measures of maximal $\mathbb{P}$ -relative entropy. In particular, when the fibre M is a surface and $h_{\text{top}}(\Theta \mid \mathbb{P}) > 0$, the set of ergodic $\mathbb{P}$ -relative measures of maximal entropy is countable. The proof refines the symbolic coding methods of [50, 21] to the skew-product setting and combines them with the finiteness techniques for factor maps due to Petersen–Quas–Shin [48]. We expect the coding-and-factor approach developed here to be useful in random dynamics, and to provide a natural quenched perspective and new tools for constructing and counting relative equilibrium states.

Structure of the paper. The paper is organised as follows. The remainder of Section 1 states the main results and presents an application. Section 2 recalls background from Pesin theory and reviews homoclinic classes. In Section 3 we construct a natural symbolic coding for the skew product Θ that preserves the base dynamics. Section 4 generalises the results of [48] to countable-state Markov shifts. Finally, Section 5 contains the proofs of the main results, Theorem A and Corollary B build upon the results established in Sections 3 and 4.

1.1. Measures of maximal relative entropy. Throughout this paper, Ω and M will always denote compact smooth (boundaryless) manifolds. Let Θ be a $\mathcal{C}^{1+\alpha}$, $\alpha > 0$, skew-product diffeomorphism

$$\begin{aligned} \Theta : \Omega \times M &\rightarrow \Omega \times M \\ (\omega, x) &\mapsto (\theta\omega, T_\omega(x)) \end{aligned} \tag{1}$$

where $\theta : \Omega \rightarrow \Omega$, $T_\omega : M \rightarrow M$ for each $\omega \in \Omega$. Moreover, we define

$$\text{proj}_\Omega : (\omega, x) \in \Omega \times M \mapsto \omega \in \Omega.$$

In this paper, all measures are implicitly assumed to be Borel measures.

This paper concerns the amount of $\mathbb{P}$ -relative entropy that Θ may admit. Below, we define the metric entropy via the Brin–Katok formula [17] since it is the shortest way to

define it (see also [57, Chapter 4]). We also define the concept of $\mathbb{P}$ -relative entropy for skew products via an extension of the Brin–Katok formula (see [61]).

Definition 1.1 (Metric entropy). Let (X, dist) be a compact metric space, $T : X \rightarrow X$ continuous, and μ a T -invariant probability measure. For $x \in X$, $\varepsilon > 0$, $n \in \mathbb{N}$, define the Bowen ball

$$B_n(x, \varepsilon) := \left\{ y \in X; \text{dist}(T^k(y), T^k(x)) < \varepsilon \text{ for } 0 \leq k < n \right\}.$$

The *metric entropy of μ with respect to T* is

$$h_\mu(T) := - \int_X \lim_{\varepsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mu(B_n(x, \varepsilon)) \, d\mu = - \int_X \lim_{\varepsilon \rightarrow 0} \liminf_{n \rightarrow \infty} \frac{1}{n} \log \mu(B_n(x, \varepsilon)) \, d\mu.$$

Definition 1.2 (Relative metric entropy). Let Θ be as (1) and μ be a Θ -invariant probability measure with $(\text{proj}_\Omega)_* \mu = \mathbb{P}$, and disintegrate $\mu(d\omega, dx) = \mu_\omega(dx) \mathbb{P}(d\omega)$. For $\omega \in \Omega$, $x \in M$, $\varepsilon > 0$, $n \in \mathbb{N}$, write

$$B_n^\omega(x, \varepsilon) := \left\{ y \in M; \text{dist}(T_\omega^k(y), T_\omega^k(x)) < \varepsilon \text{ for } 0 \leq k < n \right\},$$

where $T_\omega^k(x) := T_{\theta^k \omega} \circ T_{\theta^{k-1} \omega} \circ \dots \circ T_\omega(x)$. The $\mathbb{P}$ -relative metric entropy of μ with respect to Θ is

$$\begin{aligned} h_\mu(\Theta \mid \mathbb{P}) &:= - \int_\Omega \int_M \lim_{\varepsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log(\mu_\omega(B_n^\omega(x, \varepsilon))) \mu_\omega(dx) \mathbb{P}(d\omega) \\ &= - \int_\Omega \int_M \lim_{\varepsilon \rightarrow 0} \liminf_{n \rightarrow \infty} \frac{1}{n} \log(\mu_\omega(B_n^\omega(x, \varepsilon))) \mu_\omega(dx) \mathbb{P}(d\omega). \end{aligned}$$

We recall that if $h_\mathbb{P}(\theta) < \infty$, which will always be the case in this paper, we have that $h_\mu(\Theta \mid \mathbb{P}) = h_\mu(\Theta) - h_\mathbb{P}(\theta)$ (see [28, Chapter 5, below equation 1.1.5]).

Definition 1.3 (Measures of maximal $\mathbb{P}$ -relative entropy). Let Θ be as (1) and fix a θ -invariant probability measure $\mathbb{P}$. We define the $\mathbb{P}$ -relative topological entropy as

$$h_{\text{top}}(\Theta \mid \mathbb{P}) := \sup\{h_\nu(\Theta \mid \mathbb{P}); (\text{proj}_\Omega)_* \nu = \mathbb{P} \text{ and } \nu \text{ is a } \Theta\text{-invariant probability measure}\}.$$

We say that a Θ -invariant measure μ is a *measure of maximal $\mathbb{P}$ -relative entropy* if

$$h_\mu(\Theta \mid \mathbb{P}) = h_{\text{top}}(\Theta \mid \mathbb{P}).$$

To formulate our results, we impose additional assumptions on the skew-product base $\theta : \Omega \rightarrow \Omega$ and the measure $\mathbb{P}$, namely, we assume θ is a transitive Anosov diffeomorphism and $\mathbb{P}$ is a θ -invariant probability measure of full support, i.e.

$$\text{supp } \mathbb{P} := \{\omega \in \Omega; \mathbb{P}(U) > 0 \text{ for every open neighbourhood } U \text{ of } \omega\} = \Omega$$

Definition 1.4. Let Ω be a compact connected smooth manifold and let $\theta : \Omega \rightarrow \Omega$ be a $\mathcal{C}^{1+\alpha}$ diffeomorphism. We say that θ is *Anosov* if there exist a continuous $D\theta$ -invariant splitting $T_\omega \Omega = E^s(\omega) \oplus E^u(\omega)$, and constants $K \geq 1$, $\chi > 0$ such that for all $k \geq 0$,

$$\max \left(\left\| D\theta^k \Big|_{E^s(\omega')} \right\|, \left\| D\theta^{-k} \Big|_{E^u(\omega')} \right\| \right) \leq K e^{-\chi k}.$$

We say that θ is a *transitive Anosov* diffeomorphism if θ is Anosov and θ is topologically transitive, i.e. there exists $\omega \in \Omega$ with $\overline{\{\theta^k(\omega) : k \in \mathbb{Z}\}} = \Omega$.

1.2. Main results. Below we state our main results. The definition of hyperbolic measure used here is recalled in Definition 2.1.

Theorem A. *Let Ω and M be compact smooth manifolds, and*

$$\begin{aligned}\Theta : \Omega \times M &\rightarrow \Omega \times M \\ (\omega, x) &\mapsto (\theta\omega, T_\omega(x))\end{aligned}$$

be a $\mathcal{C}^{1+\alpha}$ diffeomorphism, for some $\alpha > 0$. Assume that $\theta : \Omega \rightarrow \Omega$ is a transitive Anosov diffeomorphism and $\mathbb{P}$ is a θ -invariant ergodic probability measure with full support. Then, Θ admits at most countably many hyperbolic ergodic measures of maximal $\mathbb{P}$ -relative entropy.

Theorem A is proved in Section 5. We mention that many natural measures satisfy the full-support assumption; for example, since θ is a transitive Anosov diffeomorphism, every ergodic equilibrium state for a Hölder potential has full support (see [16]).

We remark that Theorem A is a generalisation of the classical non-skew-product setting. Let $T : M \rightarrow M$ be a $\mathcal{C}^{1+\alpha}$ map, and $\theta : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ be Arnold's cat map

$$\theta(\omega_1, \omega_2) = (2\omega_1 + \omega_2, \omega_1 + \omega_2) \pmod{1}.$$

Consider the product map $\Theta(\omega, x) = (\theta(\omega), T(x))$ and let $\mathbb{P}$ be Lebesgue measure on $\mathbb{T}^2$, which is θ -ergodic. If $\mu = \mathbb{P} \otimes \mu'$, where μ' is a T -invariant measure of maximal entropy on M , then μ is a measure of maximal $\mathbb{P}$ -relative entropy for Θ (see [23, Fact 4.4.3]). Hence, the number of measures of maximal entropy for T is bounded above by the number of measures of maximal relative entropy over $\mathbb{P}$ for Θ . In particular, Theorem A yields the usual countability statement for ergodic hyperbolic measures of maximal entropy in the non-skew-product setting in conformance with [10, Theorem 0.5].

We stress that the hyperbolicity assumption in Theorem A is essential. Observe that the extension of θ by the identity map $\Theta(\omega, x) = (\theta(\omega), x)$ admits uncountably many ergodic measures of maximal $\mathbb{P}$ -relative entropy; in this case, all such measures have zero $\mathbb{P}$ -relative entropy. A similar example with positive $\mathbb{P}$ -relative entropy is $\Theta : \Omega \times \mathbb{T}^3 \rightarrow \Omega \times \mathbb{T}^3$ with

$$\Theta(\omega, (x_1, x_2, y)) = (\theta\omega, (2x_1 + x_2, x_1 + x_2, y)),$$

which admits uncountably many ergodic measures of maximal $\mathbb{P}$ -relative entropy.

Moreover, for fibre maps in dimension $n \geq 3$, the countability conclusion for hyperbolic measures of maximal entropy in Theorem A is sharp, i.e. there exists a $\mathcal{C}^\infty$ skew-product diffeomorphism $\Theta : \Omega \times \mathbb{T}^n \rightarrow \Omega \times \mathbb{T}^n$ satisfying the hypotheses of Theorem A which admits infinitely many hyperbolic ergodic measures of maximal $\mathbb{P}$ -relative entropy. As before, take $\mathbb{P}$ as the Lebesgue measure in $\mathbb{T}^2$ and $\theta : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ as Arnold's cat map, and set $\Theta(\omega, x) = (\theta(\omega), T(x))$, where $T : \mathbb{T}^n \rightarrow \mathbb{T}^n$ admits infinitely many hyperbolic measures of maximal entropy. One possible construction of T is as follows.

Define

$$T : (x, s) \in \mathbb{T}^{n-1} \times \mathbb{S}^1 \mapsto (A(x), F(s)) \in \mathbb{T}^{n-1} \times \mathbb{S}^1,$$

where $A \in \text{SL}(n-1, \mathbb{Z})$ is a transitive Anosov toral automorphism, and $F : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ is the function

$$F(s) = s + \delta e^{-\left(\frac{1}{s^2} + \frac{1}{(1-s)^2}\right)} \sin\left(\frac{2\pi}{s}\right) \pmod{1},$$

where $\delta > 0$ is taken sufficiently small so that F is a diffeomorphism and $F'(s) > 0$ for every $s \in \mathbb{S}^1$. One readily verifies that $F \in \mathcal{C}^\infty$ and

$$F\left(\frac{1}{j}\right) = \frac{1}{j} \text{ and } 0 < F'\left(\frac{1}{j}\right) = 1 - 2\delta\pi j^2 e^{-\left(1 + \frac{1}{(j-1)^2}\right)j^2} < 1 \text{ for every } j \in \mathbb{N}_{>1}.$$

Since circle diffeomorphisms satisfy $h_{\text{top}}(F) = 0$, we obtain that $h_{\text{top}}(T) = h_{\text{top}}(A)$. If m is the (unique) measure of maximal entropy for A (in fact, the Lebesgue measure on

$\mathbb{T}^{n-1}$), then for each $j \in \mathbb{N}_{>1}$ the product $m_j := m \otimes \delta_{1/j}$ is ergodic, hyperbolic, and satisfies $h_{m_j}(T) = h_m(A) = h_{\text{top}}(T)$. Thus T admits infinitely many hyperbolic ergodic measures of maximal entropy.

In dimension two, Theorem A can be strengthened.

Corollary B. *Under the assumptions of Theorem A if we assume that M is a compact smooth surface and $h_{\text{top}}(\Theta \mid \mathbb{P}) > 0$ then Θ admits at most countably many measures of maximal $\mathbb{P}$ -relative entropy.*

Corollary B is proved in Section 5.

In dimension two, it may be possible to strengthen the conclusion of “at most countably many measures of maximal $\mathbb{P}$ -relative entropy” of Corollary B to “at most finitely many measures of maximal $\mathbb{P}$ -relative entropy”, in analogy with the classical non-skew-product case [20, Corollary 1.4]. Another natural future direction is to derive a criterion for uniqueness. In [20] (see also [21]) it is shown that each homoclinic class supports a unique measure of maximal entropy. In particular, any smooth, transitive surface system with positive entropy has a unique measure of maximal entropy. The analogous statement for measures of maximal relative entropy remains unclear. Finally, we remark that the Anosov hypothesis may extend to Axiom A, but the symbolic system for Θ used in Theorem 3.6 would require a different construction.

It is worth mentioning that, as a consequence of a celebrated result due to Newhouse [43], $\mathcal{C}^\infty$ skew products always admit a measure of maximal $\mathbb{P}$ -relative entropy.

Proposition 1.5. *Assume that the skew product Θ is $\mathcal{C}^\infty$, and let $\mathbb{P}$ be any θ -invariant measure. Then Θ admits a $\mathbb{P}$ -relative measure of maximal entropy.*

Proof. From [43, Theorem 4.1] the map

$$\mu \in \{\nu; \nu \text{ is a } \Theta\text{-invariant probability measure}\} \mapsto h_\mu(\Theta) \in \mathbb{R}$$

is upper semi-continuous with respect to the weak* topology. Since

$$\mathcal{M}_{\mathbb{P}}(\Theta) := \{\nu; \nu \text{ is } \Theta\text{-invariant probability measure and } \nu(\text{proj}_\Omega^{-1}(\cdot)) = \mathbb{P}\}$$

is compact in the weak* topology, there exists $\tilde{\mu} \in \mathcal{M}_{\mathbb{P}}(\Theta)$ such that

$$h_{\tilde{\mu}}(\Theta) = \sup \{h_\nu(\Theta); \nu \in \mathcal{M}_{\mathbb{P}}(\Theta)\}.$$

Since $h_\nu(\Theta \mid \mathbb{P}) = h_\nu(\Theta) - h_{\mathbb{P}}(\theta)$ for each $\nu \in \mathcal{M}_{\mathbb{P}}(\Theta)$ we obtain that $\tilde{\mu}$ is a measure of maximal $\mathbb{P}$ -relative entropy. $\square$

Throughout the paper, if X is a topological space and $T : X \rightarrow X$ is a measurable map, we denote:

- $\mathcal{B}(X)$ as the Borel σ -algebra of X ;
- $\mathcal{M}(X) := \{\mu; \mu \text{ is a positive measure on } X\}$;
- $\mathcal{M}(T) := \{\mu \in \mathcal{M}(X); T_*\mu := \mu(T^{-1}(\cdot)) = \mu \text{ and } \mu \text{ is a probability measure}\}$;
- $\mathcal{M}_e(T) := \{\mu \in \mathcal{M}(T); \mu \text{ is ergodic}\}$.

1.3. An application of Corollary B. Let $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the flat torus. A classical and notoriously challenging dynamical system is the standard map on $\mathbb{T}^2$. One convenient version of the standard map is $T : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ given by

$$T(x_1, x_2) = (2x_1 - x_2 + k \cos(2\pi x_1), x_1) \pmod{1}, \text{ for some } k \in \mathbb{Z}.$$

Sinai conjectured that for k sufficiently large, T has positive metric entropy with respect to Lebesgue measure [52, Page 144]. This conjecture remains open. In contrast, much more is known about measures of maximal entropy: Obata showed that, for large k , the standard map admits a unique measure of maximal entropy; moreover, such a measure has positive metric entropy [44]. However, less is known in the random setting. Our results show that there exists at least one and, at most, countably many ergodic relative

measures of maximal entropy for certain classes of random perturbations of the standard map.

For each $k \in \mathbb{N}$, we consider the skew product

$$\begin{aligned}\Theta_k : \mathbb{T}^2 \times \mathbb{T}^2 &\rightarrow \mathbb{T}^2 \times \mathbb{T}^2 \\ (\omega, x) &\mapsto (A\omega, T_\omega(x)),\end{aligned}$$

where $A : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ is Arnold's cat map

$$A(\omega_1, \omega_2) = (2\omega_1 + \omega_2, \omega_1 + \omega_2) \pmod{1},$$

and $T_\omega : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ is a perturbed version of the standard map

$$T_\omega(x_1, x_2) := (2x_1 - x_2 + k \cos(2\pi x_1) + f(\omega), x_1) \pmod{1}, \text{ for every } \omega \in \Omega,$$

where $f : \mathbb{T}^2 \rightarrow \mathbb{S}^1$ is a C^∞ function.

Proposition 1.6. *Let $\mathbb{P}$ be a θ -invariant ergodic measure with full support. Then, there exists $k_0 \in \mathbb{N}$, such that for every $k \geq k_0$, one has $h_{\text{top}}(\Theta_k | \mathbb{P}) > 0$. In particular, Proposition 1.5 implies that Θ_k admits a measure of maximal $\mathbb{P}$ -relative entropy and Corollary B ensures that there are only countably many such ergodic measures.*

Remark 1.7. Examples of choices for f are:

- $f(\omega_1, \omega_2) = \omega_1$ (large random additive kicks);
- $f(\omega_1, \omega_2) = \varepsilon \sin(2\pi\omega_1)$ for any $\varepsilon > 0$ (arbitrarily small random additive kicks); and
- $f(\omega_1, \omega_2) = 2\omega_1 + \varepsilon \sin(2\pi\omega_2)$ (random kicks of mixed sizes depending on both base coordinates).

2. HYPERBOLIC MEASURES AND BOREL HOMOCLINIC CLASSES

In this section, we collect standard facts from hyperbolic dynamics and introduce the concept of Borel homoclinic classes (see [21, Definition 2.7]). Fix a compact smooth manifold E and a $C^{1+\alpha}$ diffeomorphism $\Theta : E \rightarrow E$, for some $\alpha > 0$. Unless stated otherwise, all statements in this section are made under this standing assumption. For the purposes of this paper, the set E can be thought of as $\Omega \times M$ as in Theorem A. Our exposition closely follows [21, Section 2].

2.1. Hyperbolic measures. By the Oseledets' multiplicative ergodic theorem [45], there exists a Borel set $E' \subset E$ such that

- E' has full measure for every Θ -invariant probability measure; and
- for every $x \in E'$ there is a finite set $\sigma(x) \subset \mathbb{R}$ and a $D\Theta$ -invariant splitting

$$T_x E = \bigoplus_{\lambda \in \sigma(x)} E_x^\lambda,$$

where each $E_x^\lambda \neq \{0\}$ is a linear subspace defined by

$$E_x^\lambda = \left\{ v \in T_x E \setminus \{0\}; \lim_{n \rightarrow \infty} \frac{1}{n} \log \|D_x \Theta^n v\| = \lambda \right\} \cup \{0\}.$$

Definition 2.1. Let $\Theta : E \rightarrow E$ be as above and $x \in E'$. The set

$$\sigma(x) = \{\lambda_1(x), \dots, \lambda_{k(x)}(x)\}$$

is called the *Lyapunov spectrum of Θ at x* , its elements are the *Lyapunov exponents of Θ at x* . For $\chi > 0$ and $\mu \in \mathcal{M}(\Theta)$ we say that μ is χ -hyperbolic if, for μ -a.e. $x \in E$.

- (i) $\min\{|\lambda|; \lambda \in \sigma(x)\} \geq \chi$;
- (ii) there are both positive and negative exponents, i.e.

$$\sigma^+(x) := \{\lambda \in \sigma(x) : \lambda > 0\} \neq \emptyset \quad \text{and} \quad \sigma^-(x) := \{\lambda \in \sigma(x) : \lambda < 0\} \neq \emptyset.$$

We then set

$$E_x^s := \bigoplus_{\lambda \in \sigma^-(x)} E_x^\lambda \text{ and } E_x^u := \bigoplus_{\lambda \in \sigma^+(x)} E_x^\lambda.$$

If $\mu \in \mathcal{M}_e(\Theta)$ is ergodic, then there exist real numbers $\lambda_1 < \dots < \lambda_k$ and integers $m_1, \dots, m_k \geq 1$ such that, for μ -a.e. $x \in E$,

$$\sigma(x) = \{\lambda_1, \dots, \lambda_k\} \text{ and } \dim E_x^{\lambda_i} = m_i, \text{ for } i = 1, \dots, k.$$

Finally, we recall that if (i) holds but (ii) fails, then μ is supported on a periodic orbit and hence has zero metric entropy [30, Corollary S.5.2] (see also the proof of [29, Theorem 4.2]).

Below, we recall a few results of Pesin theory. These definitions will be useful for defining the concept of Borel homoclinic classes in the following section.

Definition 2.2 ((χ, ε) -Pesin block). Given $\chi, \varepsilon > 0$, a (χ, ε) -Pesin block is a nonempty set $\Lambda \subset E$ for which there exists a direct sum $T_x E = E^s(x) \oplus E^u(x)$ for all $x \in \bigcup_{n \in \mathbb{Z}} \Theta^n(\Lambda)$, and a number $K > 0$ such that for any $n \in \mathbb{Z}$, $k \geq 0$, and $y \in \Lambda$,

$$\max \left(\left\| D\Theta^k \big|_{E^s(\Theta^n(y))} \right\|, \left\| D\Theta^{-k} \big|_{E^u(\Theta^n(y))} \right\| \right) \leq K e^{-\chi k + \varepsilon |n|}.$$

We recall that Pesin blocks are not necessarily Θ -invariant sets.

Below we state Pesin's stable manifold theorem, whose proof can be found at [47] (see also [21, Theorem 2.2]).

Theorem 2.3. *There exists a continuous function ε_0 with the following property: Given $\chi, \varepsilon > 0$, with $\varepsilon < \varepsilon_0(\chi)$, and a (χ, ε) -Pesin block Λ . Then each $x \in \Lambda$ admits a $\mathcal{C}^{1+\alpha}$ embedded disc $W_{\text{loc}}^s(x)$ satisfying:*

$$\text{for each } y \in W_{\text{loc}}^s(x), \limsup_{n \rightarrow \infty} \frac{1}{n} \log \text{dist}_N(\Theta^n(x), \Theta^n(y)) < 0.$$

Moreover, the embedded disc W_{loc}^s varies continuously with $x \in \Lambda$ in the $\mathcal{C}^{1+\alpha}$ topology. The same result applied to Θ^{-1} gives the local unstable manifold, which we call $W_{\text{loc}}^u(x)$.

The above theorem allows us to define the *global stable and unstable manifolds* of x lying in a (χ, ε) -Pesin block, where (χ, ε) satisfy the assumptions of Theorem 2.3, as

$$W^s(x) := \bigcup_{n \geq 0} \Theta^{-n}(W_{\text{loc}}^s(\Theta^n(x))) \text{ and } W^u(x) = \bigcup_{n \geq 0} \Theta^n(W_{\text{loc}}^s(\Theta^{-n}(x))).$$

We recall that the choice of $\varepsilon > 0$ in Theorem 2.3 does not affect the definition of the global stable and unstable manifolds [9, §8.2] (see also [21, Section 2]).

Definition 2.4. We define the *recurrent non-uniformly hyperbolic set* of Θ as

$$\text{NUH}'(\Theta) := \bigcup_{\chi > 0} \bigcap_{\varepsilon > 0} \bigcup_{\substack{(\chi, \varepsilon)\text{-Pesin} \\ \text{block } \Lambda}} \{x \in E : x \text{ is a limit of periodic points in } \Lambda\}.$$

Observe that

$$\text{NUH}'(\Theta) \subset \bigcup_{\chi > 0} \bigcap_{\varepsilon > 0} \{x \in E : x \text{ is contained in a } (\chi, \varepsilon)\text{-Pesin block}\}.$$

In particular, from Theorem 2.3 each $x \in \text{NUH}'(\Theta)$ admits global stable and unstable manifolds.

Theorem 2.5 ([29, Theorem 4.1]). *Let $\mu \in \mathcal{M}_e(\Theta)$, then $\mu(\text{NUH}'(\Theta)) = 1$.*

2.2. Borel homoclinic classes. Given two submanifolds S_1, S_2 in E we denote $S_1 \pitchfork S_2$ to be the set of transverse intersection points, i.e.

$$S_1 \pitchfork S_2 := \{x \in S_1 \cap S_2 : T_x S_1 \oplus T_x S_2 = T_x E\}.$$

Definition 2.6 ([21], Homoclinic relation). Two points $x, y \in \text{NUH}'(\Theta)$ are *homoclinically related* ($x \sim y$) if $W^s(x)$ has a transverse intersection point with an iterate of $W^u(y)$, and $W^u(x)$ has a transverse intersection point with an iterate of $W^s(y)$.

Proposition 2.7 ([21, Proposition 2.6]). *The homoclinic relation $\sim$ is an equivalence relation on $\text{NUH}'(\Theta)$.*

Definition 2.8 ([21]). *Borel homoclinic classes* are equivalence classes of $\sim$ in $\text{NUH}'(\Theta)$.

The proposition below states that the homoclinic classes characterise all the hyperbolic measures.

Proposition 2.9 ([21, Proposition 2.8]). *The following hold:*

- (i) *The set of Borel homoclinic classes is a finite or countable partition of $\text{NUH}'(\Theta)$ into invariant Borel sets;*
- (ii) *each class contains a dense set of hyperbolic periodic points;*
- (iii) *any hyperbolic ergodic measure μ is carried by a Borel homoclinic class $\mathcal{X}$; and*
- (iv) *any invariant measure carried by a Borel homoclinic class is hyperbolic.*

Remark 2.10. Note that any Borel homoclinic class that is not a singleton supports uncountably many invariant ergodic measures. If Θ admits a non-atomic hyperbolic invariant measure (i.e., not supported on a periodic orbit), then Θ admits uncountably many hyperbolic ergodic measures. Indeed, every non-trivial Borel homoclinic class contains at least one horseshoe, and each horseshoe carries uncountably many ergodic measures since it is conjugated to a full shift. By Proposition 2.9 (iv), each of these measures is hyperbolic.

We conclude this section by introducing the notions of rectangle with local product structure and the Smale bracket, which will be used throughout.

Definition 2.11 (Rectangle and Smale bracket). Let $\Theta : E \rightarrow E$ be a dynamical system on a smooth manifold E . A set $R \subset E$ is a *rectangle with local product structure* if there exist families Γ^s, Γ^u of $\mathcal{C}^1$ discs embedded in E such that:

- $R = \bigcup_{\gamma^s \in \Gamma^s} \gamma^s \cap \bigcup_{\gamma^u \in \Gamma^u} \gamma^u$;
- given $\gamma, \gamma' \in \Gamma^s$ (or Γ^u), then either $\gamma = \gamma'$ or $\gamma \cap \gamma' = \emptyset$.
- each $\gamma^s \in \Gamma^s$ is a piece of stable manifold and for each $x \in R$, there exists a unique $\gamma_x^s \in \Gamma^s$ such that $x \in \gamma_x^s$;
- each $\gamma^u \in \Gamma^u$ is a piece of unstable manifold and for each $y \in R$, there exists a unique $\gamma_y^u \in \Gamma^u$ such that $y \in \gamma_y^u$;
- for every $\gamma^s \in \Gamma^s$ and $\gamma^u \in \Gamma^u$ one has $\#(\gamma^s \cap \gamma^u) = 1$.

Given a rectangle R and $x, y \in R$, define the *Smale bracket*

$$[x, y]_\Theta := \gamma_x^s \cap \gamma_y^u,$$

which is a single point of R by the last item. We also define the sets

$$W^s(x, R) := \gamma_x^s \cap R \text{ and } W^u(x, R) := \gamma_x^u \cap R.$$

3. TOPOLOGICAL MARKOV SHIFTS

This section proves Theorem 3.6, which enables us to analyse the skew product Θ via a countable-state topological Markov shift and a factor map that preserves the base dynamics.

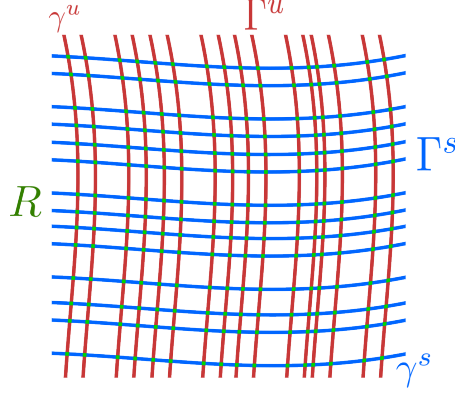


Figure 1. Illustration of a rectangle R (shown as green dots) as the intersection of segments of stable and unstable manifolds (blue and red curves, respectively).

Definition 3.1. Let S be a countable alphabet and a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, such that the set of vertices $\mathcal{V} = S$. The *topological Markov shift associated to $\mathcal{G}$* is the discrete-time topological dynamical system $\sigma : \Sigma \rightarrow \Sigma$ where

$$\Sigma = \Sigma(\mathcal{G}) = \{v = (v_i)_{i \in \mathbb{Z}} \in S^{\mathbb{Z}}; \text{ if } v_i \rightarrow v_{i+1} \in \mathcal{E} \text{ for every } i \in \mathbb{Z}\},$$

equipped with the metric $d(v, u) = \exp(-\min\{n; v_n \neq u_n\})$, and $\sigma : \Sigma \rightarrow \Sigma$ is the left shift map $\sigma((v_i)_{i \in \mathbb{Z}}) = (v_{i+1})_{i \in \mathbb{Z}}$. Given a topological Markov shift Σ associated to a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ we define the set

$$\Sigma^\# := \{(v_i)_{i \in \mathbb{Z}} \in \Sigma; \exists u', v' \in \mathcal{V} \text{ s.t. } \#\{i \in \mathbb{Z}_+; v_i = v'\} = \#\{i \in \mathbb{Z}_-; v_i = u'\} = \infty\}. \quad (2)$$

We say that Σ is irreducible if for every two symbols $v_i, v_j \in \mathcal{V}$, there exists a path in the graph $\mathcal{G}$ which connects v_i to v_j .

In the following, we recall the definition of a Markov partition.

Definition 3.2 (Markov partition). Let $\theta : \Omega \rightarrow \Omega$ be a $\mathcal{C}^{1+\alpha}$ dynamical system. A finite family $\mathcal{P} = \{P_1, \dots, P_k\}$ of rectangles with local product structure (see Definition 2.11) is a *Markov partition* if:

- (1) for each $i \in \{1, \dots, k\}$

$$P_i = \overline{\text{int}(P_i)} \text{ and } \text{int}(P_i) \cap \text{int}(P_j) = \emptyset$$

if $i \neq j$, where the interior and closure are being taken in the induced topology of $NW(\theta)$, where $NW(\theta)$ denotes the non-wandering set of $\theta : \Omega \rightarrow \Omega$;

- (2) whenever $\omega \in \text{int}(P_i)$ and $\theta(\omega) \in \text{int}(P_j)$,

$$\theta(W^s(\omega, P_i)) \subset W^s(\theta(\omega), P_j) \text{ and } \theta(W^u(\omega, P_i)) \supset W^u(\theta(\omega), P_j).$$

- (3) define the directed graph $\mathcal{G}_b = (\mathcal{V}_b, \mathcal{E}_b)$ with $\mathcal{V}_b = \mathcal{P}$ and edges

$$\mathcal{E}_b = \{P_i \rightarrow P_j; \text{int}(P_i) \cap \theta^{-1}(\text{int}(P_j)) \neq \emptyset\},$$

and let $\Sigma_b = \Sigma_b(\mathcal{G}_b)$ be the associated subshift of finite type. The coding map

$$\pi_b : (P_{i_n})_{n \in \mathbb{Z}} \mapsto \bigcap_{n \in \mathbb{Z}} \theta^{-n}(P_{i_n}) \quad (3)$$

is well defined, Hölder continuous, and satisfies the semiconjugacy $\pi_b \circ \sigma = \theta \circ \pi_b$.

- (4) the map π_b is surjective onto $NW(\theta)$ and there exists $k_{\mathcal{P}} \in \mathbb{N}$ such that $\#\pi_b^{-1}(\omega) \leq k_{\mathcal{P}}$ for every $\omega \in NW(\theta)$.

Theorem 3.3 ([15], [46, Appendix III]). *Every $\mathcal{C}^{1+\alpha}$ Axiom A diffeomorphism $\theta : \Omega \rightarrow \Omega$ admits a finite Markov partition; in particular, so does every Anosov diffeomorphism.*

Definition 3.4. Let $\theta : \Omega \rightarrow \Omega$ be an Anosov diffeomorphism and let $\mathbb{P}$ be θ -invariant ergodic probability measure on Ω . Given a Markov partition $\mathcal{P} = \{P_1, \dots, P_\kappa\}$ for θ we denote

$$\partial\mathcal{P} := \bigcup_{P \in \mathcal{P}} \partial P.$$

It is convenient to decompose the boundary into its stable and unstable parts:

$$\partial^s \mathcal{P} := \bigcup_{P \in \mathcal{P}} \partial^s P, \quad \partial^u \mathcal{P} := \bigcup_{P \in \mathcal{P}} \partial^u P,$$

where for each $P \in \mathcal{P}$,

- $\partial^s P := \{x \in P : x \notin \text{int}_{W^u(x)}(W^u(x, P))\}$; and
- $\partial^u P := \{x \in P : x \notin \text{int}_{W^s(x)}(W^s(x, P))\}$,

where interiors taken in the relative topologies of $W_{\text{loc}}^s(x)$ and $W_{\text{loc}}^u(x)$, respectively (see [14, Lemma 3.11]). Observe that $\mathbb{P}$ is ergodic, we have that $\mathbb{P}(\partial\mathcal{P}) \in \{0, 1\}$ (see [14, Proposition 3.5]).

The next statement, taken from [21, Theorem 10.5 and Section 10], is a higher-dimensional generalisation of Sarig's celebrated theorem [50, Theorem 1.3] (see also [10, Theorem 0.1]). We mention that the original result presented in [21] is more general; however, the presented version is enough for our purposes.

Theorem 3.5 ([21, Theorem 10.5 and Section 10], see also [50, Proposition 11.5] and [10, §6]). *Let $\Theta : E \rightarrow E$ be a $\mathcal{C}^{1+\alpha}$ diffeomorphism on a compact smooth manifold E . Let $\mathcal{X}$ be either a Borel homoclinic class H or E . Then, for each $\chi > 0$, there exists a countable collection of sets $\mathcal{R} = \{R_i\}_{i \in I}$, which we call a χ -Sarig partition, with the following properties:*

- Each R_i is a rectangle with local product structure (see Definition 2.11);
- whenever $x \in R_i$ and $\Theta(x) \in R_j$, then

$$\Theta(W^s(x, R_i)) \subset W^s(\Theta(x), R_j) \text{ and } \Theta(W^u(x, R_i)) \supset W^u(\Theta(x), R_j).$$

Let the topological Markov shift $\Sigma = \Sigma(\mathcal{G})$ be induced by the directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} := \mathcal{R}$ and

$$\mathcal{E} = \{R_i \rightarrow R_j; R_i \cap \Theta^{-1}(R_j) \neq \emptyset\}.$$

Then:

- (1) The map

$$\pi : (R_i)_{i \in \mathbb{Z}} \in \Sigma \mapsto \bigcap_{n \in \mathbb{N}} \overline{\bigcap_{-n \leq i \leq n} \Theta^{-i}(R_i)} \in \mathcal{X}$$

is well-defined and Hölder continuous. Moreover, $\Theta \circ \pi = \pi \circ \sigma$ and $\pi : \Sigma^\# \rightarrow \mathcal{X}$ is finite-to-one (see Definition 3.1).

- (2) Σ is a locally compact countable-state Markov shift and $\mu(\pi(\Sigma)) = \mu(\pi(\Sigma^\#)) = 1$. If $\mathcal{X}$ is a Borel homoclinic class, Σ can be chosen irreducible.
- (3) Every ergodic σ -invariant probability measure $\hat{\mu}$ on Σ projects to an ergodic Θ -invariant measure $\mu := \pi_* \hat{\mu}$ with $\mu(\mathcal{X}) = 1$ and $h_\mu(\Theta) = h_{\hat{\mu}}(\sigma)$.
- (4) For every χ -hyperbolic Θ -invariant probability measure μ on E there exists a σ -invariant probability measure $\hat{\mu}$ on Σ such that $\pi_* \hat{\mu} = \mu$.

We now state the main technical result on which our analysis is based. It refines Theorem 3.5 in the setting of skew products.

Theorem 3.6. *Let $\Theta : \Omega \times M \rightarrow \Omega \times M$ be a $\mathcal{C}^{1+\alpha}$ diffeomorphism such that*

- (a) Θ is a skew product i.e. $\Theta(\omega, x) = (\theta(\omega), T_\omega(x))$;
- (b) $\theta : \Omega \rightarrow \Omega$ is an Anosov map and $\mathbb{P}$ is a θ -invariant ergodic probability measure with full support.

Let $\mathcal{X} \subset \Omega \times M$ be equal to a Borel homoclinic class H or $\Omega \times M$. Then, for every $\chi > 0$ there exist a locally compact topological Markov shift Σ_s , a subshift of finite type Σ_b and Hölder continuous maps $\pi_s : \Sigma_s \rightarrow \mathcal{X}$ and $\pi_b : \Sigma_b \rightarrow \Omega$ such that

- (1) $\pi_s \circ \sigma_s = \Theta \circ \pi_s$ and $\pi_b \circ \sigma_b = \theta \circ \pi_b$, where σ_s is the left shift map on Σ_s and σ_b is the left shift map on Σ_b .
- (2) For every χ -hyperbolic Θ -invariant probability measure μ such that $\mu(\mathcal{X}) = 1$ with $(\text{proj}_\Omega)_* \mu = \mathbb{P}$, one has $\mu(\pi_s[\Sigma_s]) = \mu(\pi_s[\Sigma_s^\#]) = 1$;
- (3) The maps $\pi_s|_{\Sigma_s^\#} : \Sigma_s^\# \rightarrow \mathcal{X}$ and $\pi_b : \Sigma_b \rightarrow \Omega$ are finite-to-one;
- (4) There exists an 1-block map $\text{proj}_{\Sigma_b} : \Sigma_s \rightarrow \Sigma_b$ such that

$$\pi_b \circ \text{proj}_{\Sigma_b} = \text{proj}_\Omega \circ \pi_s.$$

Moreover, there exists a unique measure $\hat{\mathbb{P}}$ in $\mathcal{M}_e(\Sigma_b)$ such that $(\pi_b)_* \hat{\mathbb{P}} = \mathbb{P}$. The measure $\hat{\mathbb{P}}$; this measure also satisfies $\hat{\mathbb{P}}[\text{proj}_{\Sigma_b}(\Sigma_s)] = 1$.

- (5) If $\mathcal{X}$ is a Borel homoclinic class and there exists a χ -hyperbolic Θ -invariant ergodic probability μ with $\text{supp } \mu \subset \mathcal{X}$ and $(\text{proj}_\Omega)_* \mu = \mathbb{P}$, then the coding (Σ_s, σ_s) can be chosen to be an irreducible topological Markov shift.

Diagram 1 summarises all of the sets and projections used in the following proof.

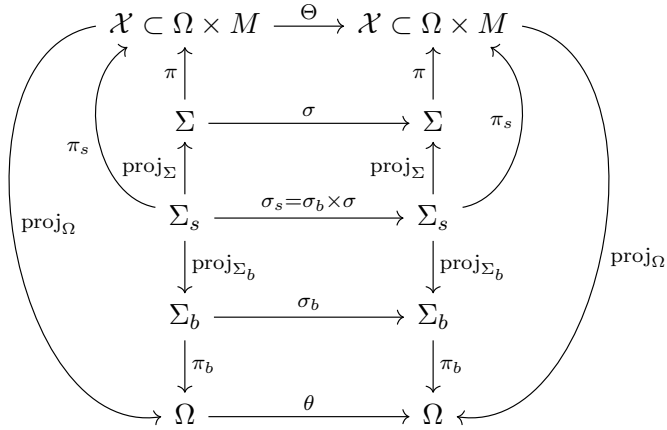


Diagram 1. Commutative diagram of the maps $\Theta, \pi, \pi_s, \pi_b, \text{proj}_\Sigma, \text{proj}_{\Sigma_b}$, and proj_Ω . We note that $\Sigma_s \subsetneq \Sigma_b \times \Sigma$, in fact, $\Sigma_s \subset \{(P, R); P \in \Sigma_b, R \in \Sigma \text{ and } \pi_b(P) = \text{proj}_\Omega \circ \pi(R)\}$.

Proof of Theorem 3.6. Let $\chi > 0$. Applying Theorem 3.5 to the skew product $\Theta : \Omega \times M \rightarrow \Omega \times M$ and $\mathcal{X}$ be either equal to Borel homoclinic class H or $\Omega \times M$, we obtain a topological Markov shift (Σ, σ) with the properties asserted therein, and let $\mathcal{R}$ be its χ -Sarig partition. We assume (Σ, σ) to be irreducible if $\mathcal{X}$ is a Borel homoclinic class. Since $\theta : \Omega \rightarrow \Omega$ is a transitive Anosov diffeomorphism and $\mathbb{P}$ is a measure with full support, there exists a subshift of finite type (Σ_b, σ_b) associated with a Markov partition $\mathcal{P}$ as in Definition 3.2. Since $\text{supp } \mathbb{P} = \Omega$ we have that $\mathbb{P}(\partial \mathcal{P}) = 0$ (see Definition 3.4). Therefore

$$\mathbb{P} \left[\omega \in \Omega; \theta^i(\omega) \in \partial \mathcal{P} \text{ for some } i \in \mathbb{Z} \right] = \mathbb{P} \left[\bigcup_{i \in \mathbb{Z}} \theta^{-i}(\partial \mathcal{P}) \right]$$

$$\leq \sum_{i \in \mathbb{Z}} \mathbb{P} [\theta^{-i}(\partial \mathcal{P})] = 0. \quad (4)$$

Let π_b be as in (3) and recall that

$$k_{\mathcal{P}} := \sup_{\omega \in \Omega} \# \pi_b^{-1}(\omega) < \infty.$$

Let $\mathcal{G}_s := (\mathcal{V}_s, \mathcal{E}_s)$ be the graph with vertices

$$\mathcal{V}_s := \{(P, R); R \in \mathcal{R}, P \in \mathcal{P} \text{ and } R \cap (\text{int } P \times M) \neq \emptyset\},$$

and edges

$$\mathcal{E}_s = \left\{ (P, R) \rightarrow (P', R') \mid \begin{array}{l} (P, R), (P', R') \in \mathcal{V}_s, \text{ and} \\ \Theta^{-1}[R' \cap (\text{int } P' \times M)] \cap R \cap (\text{int } P \times M) \neq \emptyset \end{array} \right\}.$$

Define $(\Sigma_s, \sigma_s) := (\Sigma_s(\mathcal{G}_s), \sigma_s)$. Observe that for each $(P, R) \in \mathcal{V}_s$

$$\#\{(P', R') \in \mathcal{V}_s; (P, R) \rightarrow (P', R') \in \mathcal{E}_s\} \leq \#\{R' \in \mathcal{V}_s; R \rightarrow R' \in \mathcal{E}\} \cdot \#\mathcal{P}^2 < \infty,$$

and

$$\#\{(P', R') \in \mathcal{V}_s; (P', R') \rightarrow (P, R) \in \mathcal{E}_s\} \leq \#\{R' \in \mathcal{V}_s; R' \rightarrow R \in \mathcal{E}\} \cdot \#\mathcal{P}^2 < \infty.$$

Therefore, Σ_s is a locally compact topological Markov shift.

Given $(P_i, R_i)_{i \in \mathbb{Z}} \in \Sigma_s$ we define

$$\pi_s((P_i, R_i)_{i \in \mathbb{Z}}) := \bigcap_{n \in \mathbb{N}} \overline{\bigcap_{-n \leq i \leq n} \Theta^{-i}[R_i \cap (\text{int } P_i \times M)]}.$$

We divide the remainder of the proof into seven steps.

Step 1. We show that if given $m, n \in \mathbb{Z}$ with $n > m$ and a path

$$(P_m, R_m) \rightarrow (P_{m+1}, R_{m+1}) \rightarrow \dots \rightarrow (P_n, R_n)$$

in $\mathcal{G}_s$ then

$$\bigcap_{i=m}^n \Theta^{-i}[R_i \cap (\text{int } P_i \times M)] \neq \emptyset. \quad (5)$$

In particular, we obtain that for each $(P_i, R_i)_{i \in \mathbb{Z}} \in \Sigma_s$

$$\pi_s : \Sigma_s \rightarrow \Omega \times M$$

$$(P_i, R_i)_{i \in \mathbb{Z}} \mapsto \bigcap_{n \in \mathbb{N}} \overline{\bigcap_{-n \leq i \leq n} \Theta^{-i}[R_i \cap (\text{int } P_i \times M)]}$$

is well defined and $\pi_s((P_i, R_i)_{i \in \mathbb{Z}}) = \pi((R_i)_{i \in \mathbb{Z}})$.

We follow [50, Lemma 12.1] and argue by induction on the length of the path. Fix

$$(P_m, R_m) \rightarrow (P_{m+1}, R_{m+1}) \rightarrow \dots \rightarrow (P_n, R_n) \rightarrow (P_{n+1}, R_{n+1}).$$

If $n = m + 1$ (i.e. a single edge from m to $m + 1$), the claim regarding the nonemptiness of the intersection in (5) is immediate from the definition of the edge $(P_m, R_m) \rightarrow (P_{m+1}, R_{m+1})$.

Assume the statement holds up to n , and prove it for $n + 1$. By the induction hypothesis, there exists (ω_0, x_0) such that, writing $(\omega_i, x_i) := \Theta^i(\omega_0, x_0)$,

$$(\omega_i, x_i) \in R_i \cap (\text{int } P_i \times M) \text{ for all } i = m, \dots, n.$$

Since $(P_n, R_n) \rightarrow (P_{n+1}, R_{n+1})$, choose

$$(v_n, y_n) \in (R_n \cap (\text{int } P_n \times M)) \cap \Theta^{-1}(R_{n+1} \cap (\text{int } P_{n+1} \times M)).$$

Set $(\zeta_n, z_n) := [(v_n, y_n), (\omega_n, x_n)]_\Theta \in R_n$, which is well defined because R_n is a rectangle with local product structure. Since Θ is a skew product and $(v_n, y_n), (\omega_n, x_n) \in R_n \cap (\text{int } P_n \times M)$, the bracket decouples in base and fibre:

$$[(v_n, y_n), (\omega_n, x_n)]_\Theta = ([v_n, \omega_n]_\theta, z_n),$$

and in particular $[\zeta_n, \omega_n]_\theta \in \text{int } P_n$.

Finally, define $(\zeta_0, z_0) := \Theta^{-n}(\zeta_n, z_n)$. In the following, we show that

$$(\zeta_0, z_0) \in \bigcap_{i=m}^{n+1} \Theta^{-i} [R_i \cap (\text{int } P_i \times M)],$$

which is sufficient to establish (5).

Define $(\zeta_i, z_i) := \Theta^i(\zeta_0, z_0)$ for each $i \in \{m, \dots, n\}$. For clarity in what follows, we denote by W_Θ^s (resp. W_Θ^u) the local stable (resp. unstable) manifold of Θ , and by W_θ^s (resp. W_θ^u) the local stable (resp. unstable) manifold of θ . Using the Markov properties of the partitions $\mathcal{R}$ and $\mathcal{P}$, we obtain

- $\Theta^{n+1}(\zeta_0, z_0) = \Theta(\zeta_n, z_n) \in R_{n+1} \cap (\text{int } P_{n+1} \times M)$, because

$$\Theta[W_\Theta^s((\zeta_n, z_n), R_n)] \subset W_\Theta^s(\Theta(\zeta_n, z_n), R_{n+1}) \subset R_{n+1}$$

and

$$\theta[W_\theta^s(\zeta_n, \text{int } P_n)] \subset W_\theta^s(\theta(\zeta_n), \text{int } P_{n+1}) \subset \text{int } P_{n+1}.$$

- $\Theta^n(\zeta_0, z_0) = (\zeta_n, z_n) \in R_n \cap (\text{int } P_n \times M)$ by construction.
- $\Theta^{n-1}(\zeta_0, z_0) = \Theta^{-1}(\zeta_n, z_n) \in R_{n-1} \cap (\text{int } P_{n-1} \times M)$, because

$$\Theta^n(\zeta_0, z_0) \in W_\Theta^u((\zeta_n, z_n), R_n) \subset R_n$$

hence

$$\Theta^{n-1}(\zeta_0, z_0) \in \Theta^{-1}[W_\Theta^u((\zeta_n, z_n), R_n)] \subset W_\Theta^u((\zeta_{n-1}, z_{n-1}), R_{n-1}) \subset R_{n-1},$$

and on the base,

$$\theta^n(\zeta_0) \in W_\theta^u(\zeta_n, \text{int } P_n) \subset \text{int } P_n$$

so

$$\theta^{n-1}(\zeta_0) \in \theta^{-1}[W_\theta^u(\zeta_n, \text{int } P_n)] \subset W_\theta^u(\zeta_{n-1}, \text{int } P_{n-1}) \subset \text{int } P_{n-1}.$$

- $\Theta^{n-2}(\zeta_0, z_0) \in R_{n-2} \cap (\text{int } P_{n-2} \times M)$, because from the previous item

$$\Theta^{n-1}(\zeta_0, z_0) \in W_\Theta^u((\zeta_{n-1}, z_{n-1}), R_{n-1}) \subset R_{n-1}$$

and therefore

$$\Theta^{n-2}(\zeta_0, z_0) \in \Theta^{-1}[W_\Theta^u((\zeta_{n-1}, z_{n-1}), R_{n-1})] \subset W_\Theta^u((\zeta_{n-2}, z_{n-2}), R_{n-2}) \subset R_{n-2},$$

with the analogous inclusion on the base:

$$\theta^{n-2}(\zeta_0) \in \theta^{-1}[W_\theta^u(\zeta_{n-1}, \text{int } P_{n-1})] \subset W_\theta^u(\zeta_{n-2}, \text{int } P_{n-2}) \subset \text{int } P_{n-2}.$$

Continuing in this way, we obtain

$$\Theta^{n+1-k}(\zeta_0, z_0) \in R_{n+1-k} \cap (\text{int } P_{n+1-k} \times M) \text{ for all } 0 \leq k \leq n+1-m.$$

which shows (5).

We now conclude Step 1. By (5) we have

$$\begin{aligned} \emptyset \neq \pi_s(\{(P_i, R_i)\}_{i \in \mathbb{Z}}) &= \bigcap_{n \in \mathbb{N}} \overline{\bigcap_{-n \leq i \leq n} \Theta^{-i} [R_i \cap (\text{int } P_i \times M)]} \\ &\subset \bigcap_{n \in \mathbb{N}} \overline{\bigcap_{-n \leq i \leq n} \Theta^{-i} (R_i)} = \pi((R_i)_{i \in \mathbb{Z}}). \end{aligned}$$

Since $\pi((R_i)_{i \in \mathbb{Z}})$ is a singleton, it follows that $\pi_s(\{(P_i, R_i)\}_{i \in \mathbb{Z}}) = \pi((R_i)_{i \in \mathbb{Z}})$.

Step 2. We show item (1).

Observe that the map π_s , defined above, is well defined and Hölder continuous. Indeed

$$\begin{aligned} \text{dist}_{\Omega \times M}(\pi_s(\{(P_i, R_i)\}_{i \in \mathbb{Z}}), \pi_s(\{(P'_i, R'_i)\}_{i \in \mathbb{Z}})) &\leq \text{dist}_{\Omega \times M}(\pi((R_i)_{i \in \mathbb{Z}}), \pi((R'_i)_{i \in \mathbb{Z}})) \\ &\leq C \text{dist}_{\Sigma}((R_i)_{i \in \mathbb{Z}}, (R'_i)_{i \in \mathbb{Z}})^\eta \\ &\leq C \text{dist}_{\Sigma_s}(\{(P_i, R_i)\}_{i \in \mathbb{Z}}, \{(P'_i, R'_i)\}_{i \in \mathbb{Z}})^\eta, \end{aligned}$$

where C and η are the Hölder map constants of $\pi : \Sigma \rightarrow \Omega \times M$. Moreover, it is clear from the definitions of π_s and π_b that

$$\Theta \circ \pi_s = \pi_s \circ \sigma_s \text{ and } \theta \circ \pi_b = \pi_b \circ \sigma_b.$$

Item (1) now follows.

Step 3. We show that if μ is a χ -hyperbolic, Θ -invariant measure with $\mu(\mathcal{X}) = 1$ and $(\text{proj}_\Omega)_*\mu = \mathbb{P}$, then $\mu(\pi_s(\Sigma_s)) = \mu(\pi_s(\Sigma_s^\#)) = 1$. Consequently, item (2) follows.

Recall from Theorem 3.5 (2) that if $\mu(\mathcal{X}) = 1$, then $\mu(\pi(\Sigma)) = \mu(\pi(\Sigma^\#)) = 1$ for each χ -hyperbolic Θ -invariant measure. Since $\Sigma_s^\# \subset \Sigma_s$, and $(\text{proj}_\Omega)_*\mu = \mathbb{P}$ we have from (4) that

$$\mu \left((\omega, x) \in \Omega \times M; \pi(\underline{R}) = (\omega, x) \text{ for some } \underline{R} \in \Sigma^\# \text{ and } \omega \notin \bigcup_{i \in \mathbb{Z}} \theta^{-i}(\partial \mathcal{P}) \right) = 1.$$

Therefore, it is enough to show that for each

$$\underline{R} = (R_i)_{i \in \mathbb{Z}} \in \Sigma^\# \text{ such that } \text{proj}_\Omega \circ \pi(\underline{R}) \notin \bigcup_{i \in \mathbb{Z}} \theta^{-i}(\partial \mathcal{P}),$$

there exists $\underline{P} = (P_i)_{i \in \mathbb{Z}} \in \Sigma_b$ such that $\underline{(P, R)} = \{(P_i, R_i)\}_{i \in \mathbb{Z}} \in \Sigma_s^\#$. Indeed, under this assumption, we obtain

$$\begin{aligned} \mu(\pi_s(\Sigma_s)) &\geq \mu(\pi_s(\Sigma_s^\#)) \\ &\geq \mu \left((\omega, x) \in \Omega \times M; \pi(\underline{R}) = (\omega, x) \text{ for some } \underline{R} \in \Sigma^\# \text{ and } \omega \notin \bigcup_{i \in \mathbb{Z}} \theta^{-i}(\partial \mathcal{P}) \right) \\ &= 1. \end{aligned}$$

as required.

Assume that $\underline{R} \in \Sigma^\#$ and $\pi(\underline{R}) = (\omega, x) \in (\Omega \setminus \bigcup_{i \in \mathbb{Z}} \theta^{-i}(\partial \mathcal{P})) \times M$. For each $i \in \mathbb{Z}$ let $P_i \in \mathcal{P}$ be such that $\theta^i(\omega) \in \text{int } P_i$. Since $\theta : \Omega \rightarrow \Omega$ is Anosov we have that

$$\bigcap_{-n \leq i \leq n} \theta^{-i}(\text{int } P_i) \times M = \bigcap_{-n \leq i \leq n} \Theta^{-i}(\text{int } P_i \times M)$$

is an open neighbourhood of (ω, x) in $\Omega \times M$. Since $\pi(\underline{R}) = (\omega, x)$ it follows that

$$\bigcap_{-n \leq i \leq n} \Theta^{-i}(R_i) \cap V \neq \emptyset$$

for any open neighbourhood V of (ω, x) in $\Omega \times M$. In particular,

$$\emptyset \neq \bigcap_{-n \leq i \leq n} \Theta^{-i}(R_i) \cap \bigcap_{-n \leq i \leq n} \Theta^{-i}(\text{int } P_i \times M) = \bigcap_{-n \leq i \leq n} \Theta^{-i}[R_i \cap (\text{int } P_i \times M)].$$

Therefore, we obtain that $\{(P_i, R_i)\}_{i \in \mathbb{Z}} \in \Sigma_s$. Since Σ_b admits finitely many symbols and $\underline{R} \in \Sigma^\#$, by the pigeonhole principle, it is clear that $\{(P_i, R_i)\}_{i \in \mathbb{Z}} \in \Sigma_s^\#$.

Step 4. We show that the Lipschitz 1-block map

$$\text{proj}_\Sigma : \Sigma_s \rightarrow \Sigma, \quad ((P_i, R_i)_{i \in \mathbb{Z}}) \mapsto (R_i)_{i \in \mathbb{Z}},$$

satisfies $\# \text{proj}_\Sigma^{-1}(\underline{R}) \leq k_\varnothing$ for every $\underline{R} \in \Sigma$, and item (3).

First of all, observe that given $(\underline{P}, \underline{R}), (\underline{P}', \underline{R}') \in \Sigma_s$ we have that

$$\text{dist}_\Sigma(\text{proj}_\Sigma(\underline{P}, \underline{R}), \text{proj}_\Sigma(\underline{P}', \underline{R}')) = \text{dist}_\Sigma(\underline{R}, \underline{R}') \leq \text{dist}_\Sigma(\underline{P}, \underline{P}')$$

Recall that $\pi_b : \Sigma_b \rightarrow \Omega$ is a finite-to-one surjective map over Ω . Given $(\underline{P}, \underline{R}) = (P_i, R_i)_{i \in \mathbb{Z}} \in \Sigma_s$, on the one hand we have that

$$\pi_s(\underline{P}, \underline{R}) = \bigcap_{n \in \mathbb{N}} \bigcap_{-n \leq i \leq n} \Theta^{-i} [R_i \cap (\text{int } P_i \times M)] \subset \bigcap_{i \in \mathbb{Z}} \theta^{-n}(P_n) \times M = \pi_b(\underline{P}) \times M.$$

On the other hand, we obtain that from Step 1 we have that $\pi_s(\underline{P}, \underline{R}) = \pi(\underline{R})$. It follows that

$$\begin{aligned} \#\{\text{proj}_\Sigma^{-1}(\underline{R})\} &\leq \#\left\{(\underline{P}, \underline{R}) \in \Sigma_s; \text{proj}_\Omega \circ \pi(\underline{R}) = \pi_b(\underline{P})\right\} \\ &= \sup_{\omega \in \Omega} \#\pi^{-1}(\omega) \leq \kappa_\varnothing < \infty \end{aligned}$$

We claim that $(\underline{P}, \underline{R}) \in \Sigma_s^\#$ if, and only if, $\underline{R} = \text{proj}_\Sigma((\underline{P}, \underline{R})) \in \Sigma^\#$. Indeed, if $(\underline{P}, \underline{R}) \in \Sigma_s^\#$, then by (2) there exist $(P_1, R_1), (P_2, R_2) \in \mathcal{V}_s$, such that

$$\#\{i \in \mathbb{N}; (P, R)_i = (P_1, R_1)\} = \#\{i \in \mathbb{N}; (P, R)_{-i} = (P_2, R_2)\} = \infty.$$

Therefore $\underline{R} \in \Sigma^\#$. The proof of the reverse implication can be found in the last paragraph of Step 2. Hence,

$$\#\left\{\left(\pi_s|_{\Sigma_s^\#}\right)^{-1}(\omega, x)\right\} = \#\left\{\text{proj}_\Sigma^{-1}\left((\pi|_{\Sigma^\#})^{-1}(\omega, x)\right)\right\}.$$

Since the maps proj_Σ and $\pi|_{\Sigma^\#}$ are both finite-to-one, we obtain that

$$\#\left\{\left(\pi_s|_{\Sigma_s^\#}\right)^{-1}(\omega, x)\right\} < \infty.$$

Hence $\pi_s : \Sigma_s^\# \rightarrow \mathcal{X}$ is finite-to-one. Independently, since $\pi_b : \Sigma_b \rightarrow \Omega$ is obtained from a Markov partition for θ , it is finite-to-one on Ω . This proves item (3).

Step 5. We show that for any χ -hyperbolic $\mu \in \mathcal{M}_e(\Theta)$ such that $(\text{proj}_\Omega)_*\mu = \mathbb{P}$, there exists $\hat{\mu} \in M_e(\sigma_s)$ such that $(\pi_s)_*\hat{\mu} = \mu$.

This is done exactly as in [50, Proof of Proposition 13.1]; we sketch the argument. By [50, Step 1 of Proposition 13.1] the map

$$F : (\omega, x) \in \pi_s(\Sigma_s^\#) \mapsto \#\left(\pi_s^{-1}|_{\Sigma_s^\#}\{(\omega, x)\}\right) \in \mathbb{N}$$

is measurable. Moreover, there exists $k = k_\mu$ such that $F(\omega, x) = k$ for μ -almost every $(\omega, x) \in \pi_s(\Sigma_s^\#)$ ([50, Step 1 of Proposition 13.1]; see also [60, Lemma 3.1]). Define the measure on $\Sigma_s^\#$ by

$$\hat{\mu}(E) := \int_{\Omega \times M} \frac{1}{F(\omega, x)} \left(\sum_{\pi_s(\underline{P}, \underline{R}) = (\omega, x)} \mathbb{1}_E((\underline{P}, \underline{R})) \right) d\mu(\omega, x),$$

for any Borel set $E \subset \Sigma_s^\#$. By [50, Steps 3–4 of Proposition 13.1], the measure $\hat{\mu}$ is

- a well-defined σ_s -invariant probability measure; and
- such that $(\pi_s)_*\hat{\mu} = \mu$ and $h_{\hat{\mu}}(\sigma_s) = h_\mu(\Theta)$.

Observe that, a priori, there is no need for $\hat{\mu}$ to be ergodic. The proof is finished by observing that each (typical) σ_s -invariant ergodic component $\hat{\mu}_i$ in the ergodic decomposition of $\hat{\mu}$ also satisfies $(\pi_s)_*\hat{\mu}_i = \mu$ and $h_{\hat{\mu}_i}(\sigma_s) = h_\mu(\Theta)$ (see [50, Step 5]).

Step 6. *We show item (4).*

Define

$$\text{proj}_{\Sigma_b} : \{(P_i, R_i)\}_{i \in \mathbb{Z}} \in \Sigma_s \mapsto \{P_i\}_{i \in \mathbb{Z}} \in \Sigma_b.$$

Observe that proj_{Σ_b} is a Lipschitz 1-block map; in fact:

$$\text{dist}_{\Sigma_b}(\text{proj}_{\Sigma_b}(\underline{(P, R)}), \text{proj}_{\Sigma_b}(\underline{(P', R')})) = \text{dist}_{\Sigma_b}(\underline{P}, \underline{P'}) \leq \text{dist}_{\Sigma_s}(\underline{(P, R)}, \underline{(P', R')}).$$

Let $\underline{(P, R)} \in \Sigma_s$ and $\pi_s(\underline{(P, R)}) = (\omega, x)$. Observe that, by construction, $\pi_b(\underline{P}) = \omega$. Hence

$$\text{proj}_\Omega \circ \pi_s(\underline{(P, R)}) = \text{proj}_\Omega(\omega, x) = \omega = \pi_b(\underline{P}) = \pi_b \circ \text{proj}_{\Sigma_b}(\underline{(P, R)}).$$

Observe that, by construction, $\text{proj}_{\Sigma_b} \circ \sigma_s(\underline{(P, R)}) = \sigma_b(\underline{P}) = \theta \circ \text{proj}_{\Sigma_b}(\underline{(P, R)})$.

Let μ be a χ -hyperbolic measure on $\Omega \times M$ such that $(\text{proj}_\Omega)_*\mu = \mathbb{P}$. From Step 5, there exists $\hat{\mu} \in \mathcal{M}_e(\sigma_s)$ such that $(\pi_s)_*\hat{\mu} = \mu$. We therefore obtain that

$$\mathbb{P} = (\text{proj}_\Omega)_*\mu = (\text{proj}_\Omega)_*(\pi_s)_*\hat{\mu} = (\pi_b)_*[(\text{proj}_{\Sigma_b})_*\hat{\mu}].$$

Defining the σ_b -invariant measure $\hat{\mathbb{P}} := (\text{proj}_{\Sigma_b})_*\hat{\mu}$, we have $(\pi_b)_*\hat{\mathbb{P}} = \mathbb{P}$. Since $\mathbb{P}$ has full support, (4) ensures that $\mathbb{P}[\bigcup_{i \in \mathbb{Z}} \theta^i \partial \mathcal{P}] = 0$, and therefore $\hat{\mathbb{P}}$ is the unique σ_b -invariant lift of $\mathbb{P}$ with respect to the semi-conjugacy π_b (see [14, Theorem 3.18]), which implies item (4).

Step 7. *We show item (5) and therefore conclude the proof of Theorem 3.6.*

Let $\mathcal{X}$ be a Borel homoclinic class. We restrict attention to the Borel homoclinic classes $\mathcal{X}$ with the property that there exists a χ -hyperbolic Θ -invariant ergodic probability measure μ supported on $\mathcal{X}$ such that $(\text{proj}_\Omega)_*\mu = \mathbb{P}$. Recall that $\text{supp } \mathbb{P} = \Omega$. To prove item (5) it suffices to find a subshift $\Sigma'_s \subset \Sigma_s$ such that

- Σ'_s is an irreducible topological Markov shift,
- $\text{proj}_\Sigma(\Sigma'_s) = \Sigma$, and
- $\hat{\mathbb{P}}[\text{proj}_{\Sigma_b}(\Sigma'_s)] = 1$, where $\hat{\mathbb{P}}$ is the unique measure in $\mathcal{M}_e(\Sigma_b)$ such that $(\pi_b)_*\hat{\mathbb{P}} = \mathbb{P}$.

Since $\mathcal{X}$ is a Borel homoclinic class, we may (and do) take Σ to be an irreducible two-sided topological Markov shift. Then (Σ, σ) is topologically transitive and σ is a homeomorphism. Assume Σ is infinite (the periodic-orbit case is trivial).

Because Σ is a two-sided countable Markov shift with the standard metric (see Definition 3.1), it is completely metrizable and has no isolated points. By transitivity, there exist $\underline{R}^0$ with dense forward orbit and $\underline{R}^1$ with dense backward orbit (see [49, Proposition 1.1 and footnote 5]).

By [40, Thm. 2.4] applied to $\sigma : \Sigma \rightarrow \Sigma$ and to $\sigma^{-1} : \Sigma \rightarrow \Sigma$, we obtain that the sets

$$G^0 := \left\{ \underline{Z} \in \Sigma : \underline{R}^0 \in \bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \sigma^k(\underline{Z})} \right\} \text{ and } G^1 := \left\{ \underline{Z} \in \Sigma : \underline{R}^1 \in \bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \sigma^{-k}(\underline{Z})} \right\}$$

are dense G_δ subsets of Σ . By the Baire category theorem, $G := G^0 \cap G^1$ is dense in Σ . Let $\underline{R}$ be an element of G .

Observe that the set $\bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \sigma^k(\underline{R})}$ is closed, forward σ -invariant, and it contains $\underline{R}^0$. Hence

$$\bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \sigma^k(\underline{R})} = \Sigma.$$

Applying the same argument to σ^{-1} we obtain that

$$\bigcap_{n \geq 1} \overline{\bigcup_{k \geq n} \sigma^{-k}(\underline{R})} = \Sigma.$$

Therefore every point of Σ is an accumulation point of both the forward and the backward σ -orbit of $\underline{R}$ via σ , observe that this implies that $\underline{R} \in \Sigma^\#$ since it approximates periodic points of σ .

Let $(\omega, x) := \pi(\underline{R}) \in \mathcal{X} \subset \Omega \times M$. We claim that $\omega \in \Omega \setminus (\bigcup_{i \in \mathbb{Z}} \theta^i(\partial \mathcal{P}))$. Assume, for a contradiction, that the claim is false. Then, there exists $i \in \mathbb{Z}$ with

$$\theta^i(\omega) \in \partial \mathcal{P} = \partial^s \mathcal{P} \cup \partial^u \mathcal{P}.$$

We treat the case $\theta^i(\omega) \in \partial^s \mathcal{P}$; the unstable boundary case is analogous with backward time. Since the Borel homoclinic class $\mathcal{X}$ admits a Θ -invariant ergodic measure projecting to the full-supported ergodic measure $\mathbb{P}$, by Theorem 3.5 (2) and $\mu(M \times \partial \mathcal{P}) = \mathbb{P}[\partial \mathcal{P}] = 0$ there exists $\underline{R}^* \in \Sigma$ with

$$\pi(\underline{R}^*) = (\omega^*, x^*) \text{ and } \omega^* \in \Omega \setminus \partial^s \mathcal{P}.$$

Because the forward orbit of $\underline{R}$ is dense, there is an increasing sequence $k_j \rightarrow \infty$ such that $\sigma^{k_j}(\underline{R}) \rightarrow \underline{R}^*$ as $j \rightarrow \infty$. By Theorem 3.5 (1), we have

$$\begin{aligned} \text{dist}_\Omega(\theta^{k_j}(\omega), \omega^*) &\leq \text{dist}_{\Omega \times M}(\Theta^{k_j}(\omega, x), (\omega^*, x^*)) = \text{dist}_{\Omega \times M}(\Theta^{k_j} \pi(\underline{R}), \pi(\underline{R}^*)) \\ &\leq C \text{dist}_\Sigma(\sigma^{k_j}(\underline{R}), \underline{R}^*)^\eta \rightarrow 0 \text{ as } j \rightarrow \infty \end{aligned} \quad (6)$$

for suitable constants $C, \eta > 0$.

On the other hand, by [14, Prop. 3.5],

$$\theta(\partial^s \mathcal{P}) \subset \partial^s \mathcal{P} \text{ and } \theta^{-1}(\partial^u \mathcal{P}) \subset \partial^u \mathcal{P}.$$

Hence from $\theta^i(\omega) \in \partial^s \mathcal{P}$ we get $\theta^{i+n}(\omega) \in \partial^s \mathcal{P}$ for all $n \geq 0$. In particular, for all j large enough so that $k_j \geq i$, $\theta^{k_j}(\omega) \in \partial^s \mathcal{P}$. Since $\partial^s \mathcal{P}$ is closed and $\omega^* \notin \partial^s \mathcal{P}$, the distance $\delta := \text{dist}_\Omega(\omega^*, \partial^s \mathcal{P}) > 0$, and therefore $\text{dist}_\Omega(\theta^{k_j}(\omega), \omega^*) \geq \delta$ for all such j large enough. This contradicts (6). The case $\theta^i(\omega) \in \partial^u \mathcal{P}$ is analogous. Hence $\omega \notin \bigcup_{i \in \mathbb{Z}} \theta^i(\partial \mathcal{P})$, proving the claim.

Using that $\omega \notin \bigcup_{i \in \mathbb{Z}} \theta^i(\partial \mathcal{P})$, from the proof of Step 3, there exists $(P, R) \in \Sigma_s^\#$ such that $\pi_s((P, R)) = (\omega, x)$ and $\text{proj}_\Sigma((P, R)) = \underline{R}$. Let $\mathcal{G}'_s \subset \mathcal{G}_s$ be the maximal irreducible component of $\mathcal{G}_s$ containing (P_0, R_0) (the 0-th element of (P, R)) and define $\Sigma'_s := \Sigma(\mathcal{G}'_s) \subset \Sigma(\mathcal{G}_s) = \Sigma_s$. Observe that since $(P, R) \in \Sigma_s^\#$ then $(P, R) \in \Sigma'_s$ and therefore $\sigma_s^i((P, R)) \in \Sigma'_s$ for each $i \in \mathbb{Z}$.

We claim that $\text{proj}_\Sigma(\Sigma'_s) = \Sigma$. First, density: let $U \subset \Sigma$ be nonempty open. Since the orbit of $\underline{R}$ is dense, there exists $i \in \mathbb{Z}$ with $\sigma^i(\underline{R}) \in U$. By construction, $\sigma^i(\underline{R}) = \text{proj}_\Sigma(\sigma_s^i((P, R)))$ with $\sigma_s^i((P, R)) \in \Sigma'_s$, hence $U \cap \text{proj}_\Sigma(\Sigma'_s) \neq \emptyset$. Thus $\text{proj}_\Sigma(\Sigma'_s)$ is dense.

To see that $\text{proj}_\Sigma(\Sigma'_s)$ is closed, let $\{\underline{R}^{(m)}\}_{m \geq 1} \subset \text{proj}_\Sigma(\Sigma'_s)$ with $\underline{R}^{(m)} \rightarrow \underline{R}$ in Σ as $m \rightarrow \infty$. For each m choose a lift $\underline{Y}^{(m)} = \{(P_j^{(m)}, R_j^{(m)})\}_{j \in \mathbb{Z}} \in \Sigma'_s$ such that $\text{proj}_\Sigma(\underline{Y}^{(m)}) = \underline{R}^{(m)}$ (so $R_j^{(m)} = \underline{R}_j^{(m)}$ for all j). Since Σ_b is a subshift of finite type, it is compact. Therefore, passing through a subsequence if necessary, we can assume that $\underline{P}^{(m)} := \{P_j^{(m)}\}_{j \in \mathbb{Z}} \rightarrow \underline{P} = \{P_j\}_{j \in \mathbb{Z}}$ in Σ_b as $m \rightarrow \infty$. Because $\underline{R}^{(m)} \rightarrow \underline{R}$, for each fixed j we have $R_j^{(m)} = R_j$ for all large m , and therefore,

$$\underline{Y}^{(m)} = \{(P_j^{(m)}, R_j^{(m)})\}_{j \in \mathbb{Z}} \longrightarrow \underline{Y} := \{(P_j, R_j)\}_{j \in \mathbb{Z}}.$$

The space Σ'_s is a countable-state topological Markov shift, hence closed; thus $\underline{Y} \in \Sigma'_s$. Finally, $\text{proj}_\Sigma(\underline{Y}) = \underline{R}$, so $\underline{R} \in \text{proj}_\Sigma(\Sigma'_s)$. This proves that $\text{proj}_\Sigma(\Sigma'_s)$ is closed.

In this way $\Sigma = \text{proj}_\Sigma(\Sigma'_s)$. As in Step 5, the same proof yields that $\widehat{\mathbb{P}}[\text{proj}_{\Sigma_b}(\Sigma_s)] = 1$. $\square$

4. MEASURES OF MAXIMAL RELATIVE ENTROPY ON TOPOLOGICAL MARKOV SHIFTS

This section is devoted to proving Theorem 4.4 and Corollary 4.5. Since the results are formulated in an abstract shift setting, we adopt notation consistent with the literature on relative entropy for shifts.

Definition 4.1. Given

- (1) (X, σ_X) and (Y, σ_Y) shift spaces;
- (2) $\pi : X \rightarrow Y$ a 1-block map, $\nu(\pi(X)) = 1$ and $\text{supp}(\nu) = Y$;
- (3) a σ_Y -invariant measure ν on Y ;
- (4) two σ_X -invariant measures μ_1 and μ_2 on X with $\pi_*\mu_1 = \pi_*\mu_2 = \nu$;

we define the ν -relatively independent joining of μ_1 and μ_2 as the unique measure on $X \times X$ satisfying

$$\mu_1 \otimes_\nu \mu_2(A_1 \times A_2) = \int_Y \mathbb{E}_{\mu_1}[\mathbb{1}_{A_1} \mid \pi^{-1}\mathcal{B}(Y)] \circ \pi^{-1}(y) \cdot \mathbb{E}_{\mu_2}[\mathbb{1}_{A_2} \mid \pi^{-1}\mathcal{B}(Y)] \circ \pi^{-1}(y) \nu(dy).$$

for each $A_1, A_2 \in \mathcal{B}(X)$.

In the case that (X, σ_X) is a topological Markov shift and μ is a σ_X -invariant probability measure, we denote $h_{\text{top}}(\sigma_X)$ as its Gurevich topological entropy and $h_\mu(\sigma_X)$ its metric entropy, for details see [51] (see also, [27, 26]).

Definition 4.2 ([48]). We say that μ_1 and μ_2 are ν -relatively orthogonal if

$$\mu_1 \otimes_\nu \mu_2(\{(u, v) \in X \times X; u_0 = v_0\}) = 0$$

Notation 4.3. Given a topological Markov shift (X, σ) , we denote as $\mathcal{A}_X$ its alphabet. Also, given $x \in X$ we denote as x_i its i -th coordinate. Let $a \in \mathcal{A}_X$, we define the 1-cylinder at coordinate k as $[a]_k := \{x \in X; x_k = a\}$.

The purpose of this section is to show the following theorem.

Theorem 4.4. *Let (X, σ_X) be an irreducible countable topological Markov shift with finite Gurevich topological entropy, (Y, σ_Y) an irreducible subshift of finite type and $\pi : X \rightarrow Y$ a 1-block map such that $\nu(\pi(X)) = 1$ and $\text{supp}(\nu) = Y$. Then, if μ_1 and μ_2 are ergodic measures of maximal ν -relative entropy then μ_1 and μ_2 are ν -relatively orthogonal.*

Before proving Theorem 4.4 we highlight the following useful corollary:

Corollary 4.5. *Let (X, σ_X) be an irreducible topological Markov shift with finite Gurevich topological entropy, (Y, σ_Y) an irreducible subshift of finite type and $\pi : X \rightarrow Y$ a 1-block map such that $\nu(\pi(X)) = 1$ and $\text{supp}(\nu) = Y$. Then, (X, σ_X) can only admit countably many ergodic measures of maximal ν -relative entropy.*

Proof. Assume that (X, σ_X) admits uncountably many ν -relative measures of maximal entropy $\{\mu_i\}_{i \in \mathcal{I}}$. Since $\mathcal{I}$ is uncountable while the alphabet $\mathcal{A}_X$ is countable, there exist $a \in \mathcal{A}_X$ and an uncountable subset $\mathcal{I}_0 \subset \mathcal{I}$ such that

$$\mu_i([a]_0) > 0 \text{ for every } i \in \mathcal{I}_0.$$

For each $i \in \mathcal{I}_0$ we have that

$$\mu_i([a]_0) = \int_Y \mathbb{E}_{\mu_i}[\mathbb{1}_{[a]_0} \mid \pi^{-1}\mathcal{B}(Y)] \circ \pi^{-1}(y) \nu(dy) > 0.$$

Let $g_i(y) := \mathbb{E}_{\mu_i}[\mathbb{1}_{[a]_0} \mid \pi^{-1}\mathcal{B}(Y)](\pi^{-1}(y))$; then with $A_i := \{y \in Y; g_i(y) > 0\}$ we have $\nu(A_i) > 0$.

By Theorem 4.4, for any distinct $i_1, i_2 \in \mathcal{I}_0$ the ν -relatively independent joining $\mu_{i_1} \otimes_\nu \mu_{i_2}$ gives zero mass to the coincidence event $\{(u, v) \in X \times X; u_0 = v_0\}$; in particular,

$$\mu_{i_1} \otimes_\nu \mu_{i_2}(\{(u, v); u_0 = v_0 = a\}) = 0.$$

From the definition of $\mu_{i_1} \otimes_\nu \mu_{i_2}$ given in Definition 4.1 we obtain that $\nu(A_{i_1} \cap A_{i_2}) = 0$. Hence the family $\{A_i\}_{i \in \mathcal{I}_0}$ is pairwise disjoint modulo ν -null sets.

Moreover,

$$\mathcal{I}_0 = \bigcup_{n \in \mathbb{N}} \left\{ i \in \mathcal{I}_0; \nu(A_i) > \frac{1}{n} \right\}.$$

For some n the set $\mathcal{I}_1 := \{i \in \mathcal{I}_0; \nu(A_i) > \frac{1}{n}\}$ is uncountable. But in a probability space there can be at most n pairwise disjoint measurable sets of measure $> 1/n$, a contradiction. Therefore (X, σ_X) cannot admit uncountably many ν -relative measures of maximal entropy. $\square$

From now on, we fix (X, σ_X) as an irreducible topological Markov shift with countable alphabet, (Y, σ_Y) as an irreducible subshift of finite type, and $\pi : X \rightarrow Y$ a continuous 1-block map such that $\nu(\pi(X)) = 1$.

In the following, we will construct a subshift of finite type (X_n, σ_{X_n}) which, in some sense, approximates (X, σ_X) . Since π is a 1-block map we have that $\gamma := \{\pi^{-1}([i]_0) \in \mathcal{B}(X); i \in \mathcal{A}_Y\}$ is a partition of X such that each γ is a union of 1-cylinders at coordinate 0. For each $n \in \mathbb{N}$ we define the partition of X as

$$\alpha_n := \left\{ [a_1]_0, [a_2]_0, \dots, [a_n]_0, X \setminus \left(\bigcup_{i=1}^n [a_i] \right) \right\} \vee \gamma,$$

where $\{a_i; i \in \mathbb{N}\}$ is a fixed enumeration of $\mathcal{A}_X$. Observe that each element of α_n is the union of 1-cylinders at coordinate 0.

Using the partition α_n , we define a subshift of finite type (X_n, σ_{X_n}) as follows. Let $\mathcal{G}_n = (\mathcal{V}_n, \mathcal{E}_n)$ be the directed graph with vertex set $\mathcal{V}_n := \alpha_n$ and edge set

$$\mathcal{E}_n := \{A \rightarrow A'; A, A' \in \alpha_n \text{ and } A \cap \sigma_X^{-1}(A') \neq \emptyset\}.$$

We then set $X_n := \Sigma(\mathcal{G}_n)$.

Define $\text{proj}_{X_n} : X \rightarrow X_n$ as the canonical 1-block code associated with α_n :

$$(\text{proj}_{X_n}(x))_k = A \text{ whenever } \sigma_X^k(x) \in A, A \in \alpha_n, k \in \mathbb{Z}.$$

Also, since the partition α_n refines the partition γ , for each $A \in \alpha_n$ there exists a unique $i(A) \in \{[i_0]; i_0 \in \mathcal{A}_Y\}$ such that $\pi(A) \subset [i(A)]_0 \subset Y$. This labels each atom $A \in \alpha_n$ by $i(A)$ and induces a 1-block map $\pi_n : X_n \rightarrow Y$ (induced by π defined by

$$\pi_n : (A_k)_{k \in \mathbb{N}} \in X_n \mapsto (i(A_k))_{k \in \mathbb{N}} \in Y.$$

Observe that for each $n \in \mathbb{N}$ we have that $\pi = \pi_n \circ \text{proj}_{X_n}$.

Let μ be a σ -invariant measure on X with $\pi_*\mu = \nu$, and for each $n \in \mathbb{N}$ set $\mu_n = (\text{proj}_{X_n})_*\mu$. Since $\pi = \pi_n \circ \text{proj}_{X_n}$, we have

$$(\pi_n)_*\mu_n = (\pi_n \circ \text{proj}_{X_n})_*\mu = \pi_*\mu = \nu.$$

Hence ν is supported on $\pi_n(X_n)$, i.e. $\nu(Y \setminus \pi_n(X_n)) = 0$. Because $\pi_n(X_n)$ is closed in Y and $\text{supp}(\nu) = Y$, it follows that $\pi_n(X_n) = Y$. Therefore $\pi_n : X_n \rightarrow Y$ is surjective for every $n \in \mathbb{N}$.

From the Kolmogorov-Sinai theorem (see [57, Theorem 4.9]) we have that

$$h_{\mu_n}(X_n) = H_{\mu_n} \left(\beta_n \left| \bigvee_{i=1}^{\infty} \sigma_{X_n}^{-i}(\beta_n) \right. \right) = H_{\mu} \left(\alpha_n \left| \bigvee_{i=1}^{\infty} \sigma_X^{-i}(\alpha_n) \right. \right)$$

where β_n is the partition of X_n 1-cylinders at coordinate 0. Since, $\alpha_1 \preceq \alpha_2 \preceq \dots$ and $\mathcal{B}(X) = \mathcal{B}(\bigvee_{n \in \mathbb{N}} \alpha_n)$ from [57, Theorem 4.14] (see also [56, Lemma 2.2])

$$h_\mu(\sigma_X) = \lim_{n \rightarrow \infty} h_{\mu_n}(\sigma_{X_n}).$$

With a slight abuse of notation, we write $a_1, \dots, a_n \in \mathcal{A}_{X_n}$ whenever the context is clear.

4.1. Petersen-Quas-Shin construction. Select and fix $a \in A_X$. Choose n large enough that $[a]_0 \in \alpha_n$ (so $a \in A_{X_n}$). Let μ_n^1, μ_n^2 be two distinct ergodic σ_{X_n} -invariant measures on X_n with $(\pi_n)_* \mu_n^1 = (\pi_n)_* \mu_n^2 = \nu$, and assume μ_n^1 and μ_n^2 are not ν -relatively orthogonal (Definition 4.2). Following [48], there exists an invariant measure μ_n^3 of (X_n, σ_{X_n}) with $(\pi_n)_* \mu_n^3 = \nu$ and

$$h_{\mu_n^3}(\sigma_{X_n}) > \frac{1}{2} (h_{\mu_n^1}(\sigma_{X_n}) + h_{\mu_n^2}(\sigma_{X_n})).$$

We sketch the construction of μ_n^3 . Diagram 2 summarises all semi-conjugacies used in this construction in this section. Recall, from the previous section, that $\pi_n : X \rightarrow Y$ is surjective for every $n \in \mathbb{N}$. Let $Z_n := X_n \times X_n \times \{0, 1\}^{\mathbb{Z}}$ and define the map $T := \sigma_{X_n} \times \sigma_{X_n} \times \sigma_{\text{Ber}}$, where σ_{Ber} is the shift on $\{0, 1\}^{\mathbb{Z}}$. Let ζ be the $(1/2, 1/2)$ -Bernoulli measure on $\{0, 1\}^{\mathbb{Z}}$. Also, set $\hat{\mu}_n := \mu_n^1 \otimes \mu_n^2$ and $\eta_n = \hat{\mu}_n \otimes \zeta$. Observe that η_n is T -invariant.

Define three maps $\text{proj}_{X_n}^1, \text{proj}_{X_n}^2, \text{proj}_{X_n}^3 : Z_n \rightarrow X_n$ by

- (i) $\text{proj}_{X_n}^1(u, v, r) := u$,
- (ii) $\text{proj}_{X_n}^2(u, v, r) := v$,
- (iii) for $\text{proj}_{X_n}^3$, first set, for each $k \in \mathbb{Z}$,

$$N_k^n(u, v) := \sup\{m < k; u_m = v_m = a\}, \text{ with the convention } \sup \emptyset = -\infty. \quad (7)$$

$r \in \{0, 1\}^{\mathbb{Z}}$, r_n denotes its n -th coordinate, and let

$$r \in \{0, 1\}^{\mathbb{Z}} \mapsto r_{-\infty} \in \{0, 1\} \quad (8)$$

be an independent Bernoulli($\frac{1}{2}, \frac{1}{2}$) random variable. For $k \in \mathbb{Z}$ define

$$(\text{proj}_{X_n}^3(u, v, r))_k := \begin{cases} u_k, & \text{if } r_{N_k^n(u, v)} = 0, \\ v_k, & \text{if } r_{N_k^n(u, v)} = 1. \end{cases}$$

By construction, $\text{proj}_{X_n}^3(u, v, r) \in X_n$: between successive coincidence times of u and v we copy a block from either u or v , and since X_n is a topological Markov shift, switching only at coincidence times preserves admissibility. From the definition of $\hat{\mu}_n$, we have that

$$\pi_n(u) = \pi_n(v) \text{ for } \hat{\mu}_n\text{-almost every } (u, v) \in X_n \times X_n.$$

Since π_n is 1-block map we obtain that

$$\pi_n \circ \text{proj}_{X_n}^3(z) = \pi_n \circ \text{proj}_{X_n}^2(z) = \pi_n \circ \text{proj}_{X_n}^1(z) \text{ for } \eta_n\text{-almost every } z \in Z_n. \quad (9)$$

Observe that by construction $\mu_n^1 = (\text{proj}_{X_n}^1)_* \eta_n$ and $\mu_n^2 = (\text{proj}_{X_n}^2)_* \eta_n$. We therefore define

$$\mu_n^3 = (\text{proj}_{X_n}^3)_* \eta_n,$$

Since $\text{proj}_{X_n}^3 : Z_n \rightarrow X_n$ is a 1-block factor map, μ_n^3 is σ_{X_n} -invariant.

Now, we quote a result from [48] which is contained in the proof of Theorem 1 (see [48, Equation (45)]).

Theorem 4.6 ([48]). *Under the above assumptions and notations, for any $j_* \in \mathcal{A}_{X_n}$ for each $n \in \mathbb{N}$*

$$h_{\mu_n^3}(\sigma_{X_n} \mid \nu) \geq \frac{h_{\mu_n^1}(\sigma_{X_n} \mid \nu) + h_{\mu_n^2}(\sigma_{X_n} \mid \nu)}{2} + \int_{S_n^{(a)}} \Xi_n^{j_*}(z) \eta_n(dz),$$

where

$$S_n^{(a)} := \{z \in Z_n; (\text{proj}_{X_n}^1(z))_{-1} = (\text{proj}_{X_n}^2(z))_{-1} = a\}.$$

The integrand $\Xi_n^{j*}(z)$ is given by

$$\Xi_n^{j*}(z) = \psi\left(\frac{A_n^{j*}(z) + B_n^{j*}(z)}{2}\right) - \frac{1}{2}\psi(A_n^{j*}(z)) - \frac{1}{2}\psi(B_n^{j*}(z)),$$

with $\psi(t) := -t \log t$, and

$$A_n^{j*}(z) := \mathbb{E}_{\eta_n}(\mathbb{1}_{[j*]_0} \circ \text{proj}_{X_n}^1 \mid \mathcal{B}_{1,n}^- \vee \mathcal{B}_{0,n})(z), \quad B_n^{j*}(z) := \mathbb{E}_{\eta_n}(\mathbb{1}_{[j*]_0} \circ \text{proj}_{X_n}^2 \mid \mathcal{B}_{2,n}^- \vee \mathcal{B}_{0,n})(z),$$

where

$$\mathcal{B}_{1,n}^- := \sigma(\{z \in Z_n; (\text{proj}_{X_n}^1(z))_{-m} = j\}; j \in \mathcal{A}_{X_n}, m \in \mathbb{N}),$$

$$\mathcal{B}_{2,n}^- := \sigma(\{z \in Z_n; (\text{proj}_{X_n}^2(z))_{-m} = j\}; j \in \mathcal{A}_{X_n}, m \in \mathbb{N}),$$

$$\mathcal{B}_{0,n} := (\pi_n \circ \text{proj}_{X_n}^1)^{-1}(\mathcal{B}(Y)). \quad (10)$$

Remark 4.7. Observe that by (9) one has

$$(\pi_n \circ \text{proj}_{X_n}^1)^{-1}(\mathcal{B}(Y)) = (\pi_n \circ \text{proj}_{X_n}^2)^{-1}(\mathcal{B}(Y)) = (\pi_n \circ \text{proj}_{X_n}^3)^{-1}(\mathcal{B}(Y)) \pmod{\eta_n}.$$

Thus, we may alternatively define $\mathcal{B}_{0,n} := (\pi_n \circ \text{proj}_{X_n}^i)^{-1}(\mathcal{B}(Y))$ in (10) using any $i \in \{1, 2, 3\}$, and the conclusions of Theorem 4.6 hold.

$$\begin{array}{ccccc} X \times X & & Z = X \times X \times \{0, 1\}^{\mathbb{Z}} & \xrightarrow{\text{proj}_X^i} & X \\ \downarrow p_X^i & & \downarrow \text{proj}_{Z_n} & & \downarrow \text{proj}_{X_n} \\ X & & Z_n = X_n \times X_n \times \{0, 1\}^{\mathbb{Z}} & \xrightarrow{\text{proj}_{X_n}^i} & X_n \end{array} \quad \begin{array}{c} \searrow \pi \\ Y \\ \nearrow \pi_n \end{array}$$

Diagram 2. Diagram of the maps $p_X^i, \text{proj}_{Z_n}, \text{proj}_{X_n}, \pi, \pi_n, \text{proj}_X^i$ and $\text{proj}_{X_n}^i$ with $i \in \{1, 2, 3\}$. The right diagram commutes, and all maps are semi-conjugacies between the shift maps on the respective spaces.

Sketch of the proof of Theorem 4.6. We follow closely [48, Proof of Theorem 1]. Let $\mathcal{P}$ be the partition into 0 cylinders of X_n and $\mathcal{P}_i = (\text{proj}_{X_n}^i)^{-1}\mathcal{P}$ for $i \in \{1, 2, 3\}$. For simplicity of the notation, since n is fixed we write $\mathcal{B}_1^- := \mathcal{B}_{1,n}^-, \mathcal{B}_2^- := \mathcal{B}_{2,n}^-$,

$$\mathcal{B}_3^- = \mathcal{B}_{3,n}^- = \sigma(\{z \in Z_n; (\text{proj}_{X_n}^3(z))_{-m} = j\}; j \in \mathcal{A}_{X_n} \text{ and } m \in \mathbb{N})$$

and $\mathcal{B}_{0,n} := \mathcal{B}_0$.

First of all, for each $i \in \{1, 2, 3\}$ we have that (see [48, Equation (33)])

$$h_{\mu_n^i}(\sigma_{X_n}) = H_{\eta_n}(\mathcal{P}_i \mid \mathcal{B}_i^- \vee \mathcal{B}_0) + h_{\nu}(Y).$$

By definition,

$$\begin{aligned} H_{\eta_n}(\mathcal{P}_i \mid \mathcal{B}_i^- \vee \mathcal{B}_0) &= \int_{Z_n} - \sum_{j \in \mathcal{A}_{X_n}} \mathbb{1}_{[j]_0} \circ \text{proj}_{X_n}^i \log(\mathbb{E}_{\eta_n}[\mathbb{1}_{[j]_0} \circ \text{proj}_{X_n}^i \mid \mathcal{B}_i^- \vee \mathcal{B}_0]) \, d\eta_n \\ &= \int_{Z_n} - \sum_{j \in \mathcal{A}_{X_n}} \mathbb{E}_{\eta_n}[\mathbb{1}_{[j]_0} \circ \text{proj}_{X_n}^i \mid \mathcal{B}_i^- \vee \mathcal{B}_0] \log(\mathbb{E}_{\eta_n}[\mathbb{1}_{[j]_0} \circ \text{proj}_{X_n}^i \mid \mathcal{B}_i^- \vee \mathcal{B}_0]) \, d\eta_n \end{aligned}$$

$$= \int_{Z_n} \sum_{j \in \mathcal{A}_{X_n}} \psi(\mathbb{E}_{\eta_n} [\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^i | \mathcal{B}_i^- \vee \mathcal{B}_0]) d\eta_n, \quad (11)$$

where $\psi(t) = -t \log(t)$.

Repeating verbatim the computation in [48, Proof of identity (35)], we obtain that for any $j \in \mathcal{A}_{X_n}$ the function

$$g_n^j(z) := \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^3 | \mathcal{B}_1^- \vee \mathcal{B}_2^- \vee \mathcal{B}_3^- \vee \mathcal{B}_0)(z),$$

where $z = (u, v, r)$, satisfies

- If $(\text{proj}_{X_n}^3(z))_{-1} = u_{-1} \neq v_{-1}$, then

$$g_n^j(z) = \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 | \mathcal{B}_1^- \vee \mathcal{B}_0)(z) = A_n^j(z).$$

- If $(\text{proj}_{X_n}^3(z))_{-1} = v_{-1} \neq u_{-1}$, then

$$g_n^j(z) = \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 | \mathcal{B}_2^- \vee \mathcal{B}_0)(z) = B_n^j(z).$$

- If $u_{-1} = v_{-1} = a$, then

$$\begin{aligned} g_n^j(z) &= \frac{1}{2} \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 | \mathcal{B}_1^- \vee \mathcal{B}_0)(z) + \frac{1}{2} \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 | \mathcal{B}_2^- \vee \mathcal{B}_0)(z) \\ &= \frac{1}{2} (A_n^j(z) + B_n^j(z)) \end{aligned}$$

- If $u_{-1} = v_{-1} \neq a$, then

$$g_n^j(z) = \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 | \mathcal{B}_1^- \vee \mathcal{B}_0)(z) \text{ when } r_{N_{-1}^n}(u, v) = 0$$

and

$$g_n^j(z) = \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 | \mathcal{B}_2^- \vee \mathcal{B}_0)(z) \text{ when } r_{N_{-1}^n}(u, v) = 1.$$

From (11) we obtain that

$$\begin{aligned} H_{\eta_n}(\mathcal{P}_3 | \mathcal{B}_3^- \vee \mathcal{B}_0) &\geq H_{\eta_n}(\mathcal{P}_3 | \mathcal{B}_1^- \vee \mathcal{B}_2^- \vee \mathcal{B}_3^- \vee \mathcal{B}_0) \\ &= \int_{Z_n} \sum_{j \in \mathcal{A}_{X_n}} \psi \circ g_n^j(z) d\eta_n(z) \end{aligned} \quad (12)$$

We define the sets

- $S_1 = \{z = (u, v, r) \in Z_n; (\text{proj}_{X_n}^3(z))_{-1} = u_{-1} \neq v_{-1}\};$
- $S_2 = \{z = (u, v, r) \in Z_n; (\text{proj}_{X_n}^3(z))_{-1} = v_{-1} \neq u_{-1}\};$
- $S_3^{(a)} = \{z = (u, v, r) \in Z_n; u_{-1} = v_{-1} = a\};$ and
- $S_3^{(\neq a)} = \{z = (u, v, r) \in Z_n; u_{-1} = v_{-1} \neq a\}.$

As in [48, Equations (41), (42) and (43)] we have that

$$\begin{aligned} \int_{S_1 \cup S_2} \psi \circ g_n^j(z) d\eta_n(z) &= \frac{1}{2} \int_{S_1 \cup S_2} \psi(\mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 | \mathcal{B}_1^- \vee \mathcal{B}_0))(z) d\eta_n(z) \\ &\quad + \frac{1}{2} \int_{S_1 \cup S_2} \psi(\mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 | \mathcal{B}_2^- \vee \mathcal{B}_0))(z) d\eta_n(z) \end{aligned} \quad (13)$$

Since $\eta_n = \mu_n^1 \otimes_\nu \mu_n^2 \otimes \zeta$ and ζ is $(\frac{1}{2}, \frac{1}{2})$ -Bernoulli we have that

$$\zeta(r_{N_{-1}^n}(u, v) = 0) = \zeta(r_{N_{-1}^n}(u, v) = 1) = \frac{1}{2}$$

and we obtain that

$$\int_{S_3^{(\neq a)}} \psi \circ g_n^j d\eta_n = \frac{1}{2} \int_{S_3^{(\neq a)}} \psi \circ \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 | \mathcal{B}_1^- \vee \mathcal{B}_0) d\eta_n$$

$$+ \frac{1}{2} \int_{S_3^{(\neq a)}} \psi \circ \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 \mid \mathcal{B}_2^- \vee \mathcal{B}_0) d\eta_n. \quad (14)$$

Furthermore,

$$\begin{aligned} \int_{S_3^{(a)}} \psi \circ g_n^j(z) d\eta_n &= \int_{S_3^{(a)}} \psi \left(\frac{1}{2} (A_n^j + B_n^j) \right) d\eta_n \\ &= \int_{S_3^{(a)}} \frac{1}{2} \psi \circ A_n^j d\eta_n + \int_{S_3^{(a)}} \frac{1}{2} \psi \circ B_n^j d\eta_n + \int_{S_3^{(a)}} \Xi_n^j d\eta_n. \end{aligned} \quad (15)$$

Combining (12), (13), (14) and (15) we obtain that

$$\begin{aligned} h_{\mu_3^n}(\sigma_{X_n} \mid \nu) &= H_{\eta_n}(\mathcal{P}_3 \mid \mathcal{B}_3^- \vee \mathcal{B}_0) \geq \int_{Z_n} \sum_{j \in \mathcal{A}_{X_n}} \psi \circ g_n^j d\eta_n \\ &= \int_{S_1 \cup S_3 \cup S_3^{(\neq a)}} \sum_{j \in \mathcal{A}_{X_n}} \psi \circ g_n^j d\eta_n + \int_{S_3^{(a)}} \sum_{j \in \mathcal{A}_{X_n}} \psi \circ g_n^j d\eta_n \\ &\geq \frac{1}{2} \int_{X_n} \psi \circ \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^1 \mid \mathcal{B}_1^- \vee \mathcal{B}_0) d\eta_n \\ &\quad + \frac{1}{2} \int_{X_n} \psi \circ \mathbb{E}_{\eta_n}(\mathbf{1}_{[j]_0} \circ \text{proj}_{X_n}^2 \mid \mathcal{B}_2^- \vee \mathcal{B}_0) d\eta_n + \sum_{j \in \mathcal{A}_{X_n}} \int_{S_3^{(a)}} \Xi_n^j d\eta_n \\ &\geq \frac{h_{\mu_1^n}(\sigma_{X_n} \mid \nu) + h_{\mu_2^n}(\sigma_{X_n} \mid \nu)}{2} + \int_{S_3^{(a)}} \Xi_n^{j^*} d\eta_n. \end{aligned}$$

□

Below we quote another result from [48]. In [48], the statement is proved under the assumption that X is a subshift of finite type, but the same proof works assuming that X is a countable-state irreducible Markov shift and Y is a subshift of finite type.

Lemma 4.8 ([48, Lemma 3]). *Consider a two-sided countable Markov shift $X \subset \mathcal{A}_X^{\mathbb{Z}}$ with shift σ_X , a subshift Y with shift σ_Y , and a 1-block map $\pi : X \rightarrow Y$ such that $\nu(\pi(X)) = 1$ and $\text{supp}(\nu) = Y$. I have added the following sentence in red. Let μ_1, μ_2 be σ_X -ergodic invariant measures satisfying $\pi_*(\mu_1) = \pi_*(\mu_2) = \nu$. Denote by $\hat{\mu} := \mu_1 \otimes_{\nu} \mu_2$ their relatively independent joining on $X \times X$, and by $p_X^1, p_X^2 : X \times X \rightarrow X$ the coordinate projections.*

Put $\mathcal{F}^- := \sigma(x_k; k < 0)$ for the past σ -algebra on X , and set $\mathcal{F}_i^- := (p_X^i)^{-1}(\mathcal{F}^-)$, $i \in \{1, 2\}$ on $X \times X$. Write $\mathcal{Q}_Y := \sigma([y_0]_0; y_0 \in \mathcal{A}_Y)$ and define

$$\mathcal{F}_0 := (\pi \circ p_X^1)^{-1}(\mathcal{Q}_Y) = (\pi \circ p_X^2)^{-1}(\mathcal{Q}_Y) \pmod{\hat{\mu}}.$$

Assume that $\tilde{S} := \{(u, v) \in X \times X; u_{-1} = v_{-1}\}$ has positive $\hat{\mu}$ -measure, and

$$\mathbb{E}_{\hat{\mu}}(\mathbf{1}_{[j]_0} \circ \text{proj}_X^1 \mid \mathcal{F}_1^- \vee \mathcal{F}_0) = \mathbb{E}_{\hat{\mu}}(\mathbf{1}_{[j]_0} \circ \text{proj}_X^2 \mid \mathcal{F}_2^- \vee \mathcal{F}_0) \quad \hat{\mu}\text{-a.e. on } \tilde{S},$$

for every $j \in \mathcal{A}_X$, then $\mu_1 = \mu_2$. In particular there exists $a, b \in \mathcal{A}_X$ such that

$$\hat{\mu} \left\{ (u, v) \in X \times X \left| \begin{array}{l} u_{-1} = v_{-1} = a, \text{ and} \\ \mathbb{E}_{\hat{\mu}}(\mathbf{1}_{[b]_0} \circ \text{proj}_X^1 \mid \mathcal{F}_1^- \vee \mathcal{F}_0) \neq \mathbb{E}_{\hat{\mu}}(\mathbf{1}_{[b]_0} \circ \text{proj}_X^2 \mid \mathcal{F}_2^- \vee \mathcal{F}_0) \end{array} \right. \right\} > 0$$

Proof of Theorem 4.4. Let $a, b \in \mathcal{A}_X$ be as given at the end of Lemma 4.8, and let $n \in \mathbb{N}$ be large enough such that $[a]_0, [b]_0 \in \alpha_n$; in this way we have that $a, b \in \mathcal{A}_{X_n}$. We assume the notations of Theorem 4.6 and Lemma 4.8. Suppose, for contradiction, that the theorem is false. Then there exist two ergodic measures of maximal ν -relative entropy, μ^1 and μ^2 , such that

$$\mu^1 \otimes_{\nu} \mu^2(\{(u, v) \in X \times X; u_0 = v_0\}) > 0.$$

Let $\eta := \mu^1 \otimes_\nu \mu^2 \otimes \zeta$, recalling that ζ is the Bernoulli($\frac{1}{2}, \frac{1}{2}$) probability measure on $\{0, 1\}^\mathbb{Z}$. Define $\mu_3 = (\text{proj}_X^3)_* \eta$, where for $k \in \mathbb{Z}$ we set

$$N_k(u, v) := \sup\{m < k; u_m = v_m = a\} \text{ with } \sup \emptyset = -\infty.$$

Recalling the definition of $r_{-\infty}$ in (8) we define

$$(\text{proj}_X^3(u, v, r))_k := \begin{cases} u_k, & \text{if } r_{N_k(u, v)} = 0, \\ v_k, & \text{if } r_{N_k(u, v)} = 1. \end{cases}$$

Observe that proj_X^3 is the non-truncated analogue of the $\text{proj}_{X_n}^3$ function in Section 4.1, obtained by replacing X_n with X . We will show that

$$h_{\mu^3}(X) > h_{\mu^1}(\sigma_X) = h_{\mu^2}(\sigma_X) = h_{\text{top}}(\sigma_X | \nu).$$

which yields a contradiction. Note that we cannot directly apply the result of Lemma 4.8 since (X, σ_X) is not necessarily a subshift of finite type.

We define $\mu_n^i = (\text{proj}_{X_n})_* \mu^i$ for $i \in \{1, 2\}$ and we recall that $\eta_n = \mu_1^n \otimes_\nu \mu_2^n \otimes \zeta$. Let μ_n^3 be the σ_{X_n} -invariant probability measure constructed in Section 4.1. Observe that by our choice of $n \in \mathbb{N}$ the symbol $a \in \mathcal{A}_X$ also lies in the alphabet of its truncation $\mathcal{A}_{X_n}$.

We claim that $(\text{proj}_{X_n})_* \mu^3 = \mu_n^3$, which is equivalent (from the definitions of μ^3 and μ_n^3) to showing that

$$\begin{aligned} (\text{proj}_{X_n})_* (\text{proj}_X^3)_* (\mu^1 \otimes_\nu \mu^2 \otimes \zeta) &= (\text{proj}_{X_n}^3)_* [(\text{proj}_{X_n})_* \mu^1 \otimes_\nu (\text{proj}_{X_n})_* (\mu^2 \otimes \zeta)] \\ &= (\text{proj}_{X_n}^3)_* (\mu_n^1 \otimes_\nu \mu_n^2 \otimes \zeta) \end{aligned} \quad (16)$$

To show (16), observe from (7) and the fact that $a \in \mathcal{A}_{X_n}$ we have that

$$\begin{aligned} N_k(u, v) &= \sup\{m < k; u_m = v_m = a\} = \sup\{m < k; \text{proj}_{X_n}(u)_m = \text{proj}_{X_n}(v)_m = a\} \\ &= N_k^n(\text{proj}_{X_n}(u), \text{proj}_{X_n}(v)) \end{aligned} \quad (17)$$

From (17), we obtain that for every $k \in \mathbb{Z}$

$$\begin{aligned} (\text{proj}_{X_n} \circ \text{proj}_X^3(u, v, r))_k &= \begin{cases} \text{proj}_{X_n}(u)_k, & \text{if } r_{N_k(u, v)} = 0, \\ \text{proj}_{X_n}(v)_k, & \text{if } r_{N_k(u, v)} = 1. \end{cases} \\ &= \begin{cases} \text{proj}_{X_n}(u)_k, & \text{if } r_{N_k^n(\text{proj}_{X_n}(u), \text{proj}_{X_n}(v))} = 0, \\ \text{proj}_{X_n}(v)_k, & \text{if } r_{N_k^n(\text{proj}_{X_n}(u), \text{proj}_{X_n}(v))} = 1. \end{cases} \\ &= \text{proj}_{X_n}^3(\text{proj}_{X_n}(u), \text{proj}_{X_n}(v), r) \end{aligned}$$

Which implies (16) and therefore that $\text{proj}_{X_n} \mu^3 = \mu_n^3$

In this way, we have that

$$h_{\mu^i}(X) = \lim_{n \rightarrow \infty} h_{\mu_n^i}(X_n) \text{ for } i \in \{1, 2, 3\}, \text{ where } \mu_n^i := (\text{proj}_{X_n})_* \mu^i.$$

By Lemma 4.8 there exists $b \in \mathcal{A}_X$ (which also lies in $\mathcal{A}_{X_n}$ for n large enough) such that, on a $\mu^1 \otimes_\nu \mu^2$ -positive subset of $\{(u, v) \in X \times X : u_{-1} = v_{-1} = a\}$,

$$\mathbb{E}_{\mu^1 \otimes_\nu \mu^2}(\mathbb{1}_{[b]_0} \circ p_X^1 \mid \mathcal{F}_1^- \vee \mathcal{B}_0) \neq \mathbb{E}_{\mu^1 \otimes_\nu \mu^2}(\mathbb{1}_{[b]_0} \circ p_X^2 \mid \mathcal{F}_2^- \vee \mathcal{B}_0).$$

Hence

$$\mathbb{E}_\eta(\mathbb{1}_{[b]_0} \circ \text{proj}_X^1 \mid \mathcal{B}_1^- \vee \mathcal{B}_0) \neq \mathbb{E}_\eta(\mathbb{1}_{[b]_0} \circ \text{proj}_X^2 \mid \mathcal{B}_2^- \vee \mathcal{B}_0) \quad (18)$$

on an η -positive measure set contained in

$$S^{(a)} := \{z \in Z; (\text{proj}_X^1(z))_{-1} = (\text{proj}_X^2(z))_{-1} = a\}.$$

Let $\text{proj}_{Z_n} : Z \rightarrow Z_n$ be the projection

$$\text{proj}_{Z_n} : Z \rightarrow Z_n$$

$$z = (x, y, r) \mapsto (\text{proj}_{X_n}(x), \text{proj}_{X_n}(y), r),$$

and observe that

$$\text{proj}_{X_n}^i \circ \text{proj}_{Z_n} = \text{proj}_{X_n} \circ \text{proj}_X^i \text{ for any } i \in \{1, 2, 3\}.$$

For each $i \in \{1, 2\}$ the sequence of σ -algebras

$$(\text{proj}_{Z_n})^{-1}(\mathcal{B}_{i,n}^- \vee \mathcal{B}_{0,n}), \quad n \geq 1,$$

is increasing and

$$\sigma \left(\bigvee_{n \geq 1} (\text{proj}_{Z_n})^{-1}(\mathcal{B}_{i,n}^- \vee \mathcal{B}_{0,n}) \right) = \mathcal{B}_i^- \vee \mathcal{B}_0.$$

By Doob's martingale convergence theorem,

$$\mathbb{E}_\eta(\mathbb{1}_{[b]_0} \circ \text{proj}_X^i \mid \mathcal{B}_i^- \vee \mathcal{B}_0) = \lim_{n \rightarrow \infty} \mathbb{E}_\eta \left(\mathbb{1}_{[b]_0} \circ \text{proj}_X^i \mid (\text{proj}_{Z_n})^{-1}(\mathcal{B}_{i,n}^- \vee \mathcal{B}_{0,n}) \right) \quad (19)$$

in $L^1(Z, \eta)$ and η -a.s. Moreover,

$$\text{proj}_{Z_n}(S^{(a)}) \subset S_n^{(a)} := \{z \in Z_n; (\text{proj}_{X_n}^1(z))_{-1} = (\text{proj}_{X_n}^2(z))_{-1} = a\}.$$

With $\psi(t) := -t \log t$,

$$\begin{aligned} & \int_{S_n^{(a)}} \psi \left(\frac{A_n^b + B_n^b}{2} \right) - \frac{1}{2} \psi(A_n^b) - \frac{1}{2} \psi(B_n^b) d\eta_n \\ &= \int_{\text{proj}_{Z_n}^{-1}(S_n^{(a)})} \psi \left(\frac{A_n^b \circ \text{proj}_{Z_n} + B_n^b \circ \text{proj}_{Z_n}}{2} \right) - \frac{1}{2} \psi(A_n^b \circ \text{proj}_{Z_n}) - \frac{1}{2} \psi(B_n^b \circ \text{proj}_{Z_n}) d\eta \\ &\geq \int_{S^{(a)}} \psi \left(\frac{A_n^b \circ \text{proj}_{Z_n} + B_n^b \circ \text{proj}_{Z_n}}{2} \right) - \frac{1}{2} \psi(A_n^b \circ \text{proj}_{Z_n}) - \frac{1}{2} \psi(B_n^b \circ \text{proj}_{Z_n}) d\eta, \quad (20) \end{aligned}$$

where A_n^b, B_n^b are as in Theorem 4.6 taking $j_* = b$. Observe that

$$\begin{aligned} A_n^b \circ \text{proj}_{Z_n} &= \mathbb{E}_{\eta_n}(\mathbb{1}_{[b]_0} \circ \text{proj}_{X_n}^1 \mid \mathcal{B}_{1,n}^- \vee \mathcal{B}_{0,n}) \circ \text{proj}_{Z_n} \\ &= \mathbb{E}_\eta(\mathbb{1}_{[b]_0} \circ \text{proj}_X^1 \mid (\text{proj}_{Z_n})^{-1}(\mathcal{B}_{1,n}^- \vee \mathcal{B}_{0,n})) \end{aligned} \quad (21)$$

Similarly, we obtain that

$$B_n^b \circ \text{proj}_{Z_n} = \mathbb{E}_\eta(\mathbb{1}_{[b]_0} \circ \text{proj}_X^2 \mid (\text{proj}_{Z_n})^{-1}(\mathcal{B}_{2,n}^- \vee \mathcal{B}_{0,n})). \quad (22)$$

Combining (20) with (21), (22), (18), and (19) yields

$$\lim_{n \rightarrow \infty} \int_{S_n^{(a)}} \left[\psi \left(\frac{A_n^b + B_n^b}{2} \right) - \frac{1}{2} \psi(A_n^b) - \frac{1}{2} \psi(B_n^b) \right] d\eta_n > 0. \quad (23)$$

Finally, Theorem 4.6 together with (23) gives

$$\begin{aligned} h_{\mu^3}(\sigma_X \mid \nu) &= \lim_{n \rightarrow \infty} h_{\mu_n^3}(\sigma_{X_n} \mid \nu) \\ &\geq \lim_{n \rightarrow \infty} \left[\frac{h_{\mu_n^1}(\sigma_{X_n} \mid \nu) + h_{\mu_n^2}(\sigma_{X_n} \mid \nu)}{2} + \int_{S_n^{(a)}} \Xi_n(z) \eta_n(dz) \right] \\ &\geq \lim_{n \rightarrow \infty} \left[\frac{h_{\mu_n^1}(\sigma_{X_n} \mid \nu) + h_{\mu_n^2}(\sigma_{X_n} \mid \nu)}{2} + \int_{S_n^{(a)}} \psi \left(\frac{A_n^b + B_n^b}{2} \right) - \frac{1}{2} \psi(A_n^b) - \frac{1}{2} \psi(B_n^b) d\eta_n \right] \\ &> \frac{h_{\mu^1}(\sigma_X \mid \nu) + h_{\mu^2}(\sigma_X \mid \nu)}{2} = h_{\mu^1}(\sigma_X \mid \nu). \end{aligned}$$

This contradicts $h_{\mu^1}(\sigma_X \mid \nu) = h_{\text{top}}(\sigma_X \mid \nu)$, since $\pi_* \mu^3 = \nu$. $\square$

5. PROOFS OF THEOREM A, COROLLARY B AND PROPOSITION 1.6

Before proving Theorem A, we recall a result from the literature and prove a lemma. The proof of the Lemma below was sketched in Step 5 of the proof of Theorem 3.6.

Lemma 5.1 ([50, Proposition 13.2 and Step 2 of its proof]). *Let Θ be as in Theorem A and fix $\chi > 0$. Consider Σ_s and $\pi_s : \Sigma_s \rightarrow \Omega \times M$ as in Theorem 3.6. Then, for each χ -hyperbolic $\mu \in \mathcal{M}_e(\Theta)$*

- (i) *there exists $\hat{\mu} \in M_e(\sigma_s)$ such that $(\pi_s)_*\hat{\mu} = \mu$;*
- (ii) *$h_{\hat{\mu}}(\sigma_s) = h_{\mu}(\Theta)$; and*
- (iii) *there exists N_{μ} such that $\#\pi_s^{-1}(\omega, x) = N_{\mu}$ for μ -almost every $(\omega, x) \in \Omega \times M$.*

Lemma 5.2. *Let Θ and $\mathbb{P}$ be as in Theorem A. Define*

$$\mathcal{M}_{\max}(\Theta \mid \mathbb{P}) := \left\{ \mu \in \mathcal{M}_e(\Theta) \left| \begin{array}{l} (\text{proj}_{\Omega})_*\mu = \mathbb{P} \text{ and } \mu \text{ is a hyperbolic} \\ \text{measure of } \mathbb{P}\text{-maximal relative entropy} \end{array} \right. \right\},$$

If, for each $\chi > 0$ and every homoclinic class $\mathcal{X}$, the set

$$\mathcal{M}_{\max}^{\chi, \chi}(\Theta \mid \mathbb{P}) := \left\{ \mu \in \mathcal{M}_e(\Theta) \left| \begin{array}{l} (\text{proj}_{\Omega})_*\mu = \mathbb{P}, \mu \text{ is } \chi\text{-hyperbolic}, \mu(\mathcal{X}) = 1, \text{ and} \\ \mu \text{ is a hyperbolic measure of } \mathbb{P}\text{-maximal relative entropy} \end{array} \right. \right\}, \quad (24)$$

is countable, then $\mathcal{M}_{\max}(\Theta \mid \mathbb{P})$ is countable.

Proof. Define $\mathcal{M}_{\max}^{\chi}(\Theta \mid \mathbb{P})$ as in (24) but changing $\mathcal{X}$ to $\Omega \times M$. Observe that

$$\mathcal{M}_{\max}(\Theta \mid \mathbb{P}) = \bigcup_{\chi > 0} \mathcal{M}_{\max}^{\chi}(\Theta \mid \mathbb{P}) = \bigcup_{n \in \mathbb{N}} \mathcal{M}_{\max}^{\frac{1}{n}}(\Theta \mid \mathbb{P}).$$

Thus it suffices to show that each $\mathcal{M}_{\max}^{\chi}(\Theta \mid \mathbb{P})$ is countable.

Fix $\chi > 0$. Let $\mu \in \mathcal{M}_e(\Theta)$ be χ -hyperbolic. By Theorem 2.5, $\mu(\text{NUH}'(\Theta)) = 1$. From Proposition 2.9, there exists a countable family of Borel homoclinic classes $\{\mathcal{X}_i\}_{i \in \mathcal{I}}$ forming a pairwise disjoint, Θ -invariant partition of $\text{NUH}'(\Theta)$. Since μ is ergodic, there is a unique $i_0 \in \mathcal{I}$ such that $\mu(\mathcal{X}_{i_0}) = 1$. Consequently,

$$\mathcal{M}_{\max}^{\chi}(\Theta \mid \mathbb{P}) = \bigcup \{ \mathcal{M}_{\max}^{\chi, \chi}(\Theta \mid \mathbb{P}) ; \mathcal{X} \text{ is a Borel homoclinic class} \} = \bigcup_{i \in \mathcal{I}} \mathcal{M}_{\max}^{\chi, \chi}(\Theta \mid \mathbb{P}).$$

By hypothesis, each $\mathcal{M}_{\max}^{\chi, \chi}(\Theta \mid \mathbb{P})$ is countable, and $\mathcal{I}$ is countable; hence $\mathcal{M}_{\max}^{\chi}(\Theta \mid \mathbb{P})$ is countable. Taking the countable union over $n \geq 1$ gives that $\mathcal{M}_{\max}(\Theta \mid \mathbb{P})$ is countable. $\square$

Now we can prove Theorem A.

Proof of Theorem A. Let $\mathbb{P}$ be a θ -invariant ergodic probability measure with full support. By Lemma 5.2, it suffices to show that $\mathcal{M}_{\max}^{\chi, \chi}(\Theta \mid \mathbb{P})$ is countable for each homoclinic class $\mathcal{X}$ and each $\chi > 0$.

Fix $\chi > 0$ and a Borel homoclinic class $\mathcal{X}$. Let Σ_s be an irreducible locally compact topological Markov shift, Σ_b a subshift of finite type, and let $\pi_s : \Sigma_s \rightarrow \mathcal{X}$ and $\pi_b : \Sigma_b \rightarrow \Omega$ be the Hölder maps given by Theorem 3.6. Recall that:

- (1) $\pi_s \circ \sigma_s = \Theta \circ \pi_s$ and $\pi_b \circ \sigma_b = \theta \circ \pi_b$, where σ_s and σ_b denote the left shift maps on Σ_s and Σ_b , respectively;
- (2) $\pi_s[\Sigma_s^{\#}]$ has full measure for every χ -hyperbolic $\mu \in \mathcal{M}_e(\Theta)$ with $\mu(\mathcal{X}) = 1$; moreover $\pi_s : \Sigma_s^{\#} \rightarrow \mathcal{X}$ is finite-to-one, and $\pi_b : \Sigma_b \rightarrow \Omega$ is a surjective finite-to-one map;

(3) there exists a 1-block map $\text{proj}_{\Sigma_b} : \Sigma_s \rightarrow \Sigma_b$ such that

$$\pi_b \circ \text{proj}_{\Sigma_b} = \text{proj}_{\Omega} \circ \pi_s,$$

with $\widehat{\mathbb{P}}[\text{proj}_{\Sigma_b}(\Sigma_s)] = 1$, where $\widehat{\mathbb{P}}$ unique measure in $M_e(\sigma_b)$ such that $(\pi_b)_*\widehat{\mathbb{P}} = \mathbb{P}$.

Let $\mu \in \mathcal{M}_e(\Theta)$ be a hyperbolic measure of $\mathbb{P}$ -maximal relative entropy. By Lemma 5.1, there exists $\widehat{\mu} \in \mathcal{M}_e(\sigma_s)$ such that $(\pi_s)_*\widehat{\mu} = \mu$ and $h_\mu(\Theta) = h_{\widehat{\mu}}(\sigma_s)$. Observe that

$$(\pi_b)_*[(\text{proj}_{\Sigma_b})_*\widehat{\mu}] = (\pi_b \circ \text{proj}_{\Sigma_b})_*\widehat{\mu} = (\text{proj}_{\Omega})_*(\pi_s)_*\widehat{\mu} = (\text{proj}_{\Omega})_*\mu = \mathbb{P}.$$

Thus $(\text{proj}_{\Sigma_b})_*\widehat{\mu} = \widehat{\mathbb{P}}$, since the lift of $\mathbb{P}$ via π_b is unique (recall that $\mathbb{P}$ is assumed to be fully supported on Ω).

Claim. For the factor map $\text{proj}_{\Sigma_b} : (\Sigma_s, \sigma_s) \rightarrow (\Sigma_b, \sigma_b)$, $\widehat{\mu}$ is a measure of maximal $\widehat{\mathbb{P}}$ -relative entropy.

Proof of the claim. Assume that the claim is false for a contradiction. Then, there exists $\widehat{\eta} \in \mathcal{M}_e(\sigma_s)$ with $(\text{proj}_{\Sigma_b})_*\widehat{\eta} = \widehat{\mathbb{P}}$ such that $h_{\widehat{\eta}}(\sigma_s | \widehat{\mathbb{P}}) > h_{\widehat{\mu}}(\sigma_s | \widehat{\mathbb{P}})$. Set $\eta := (\pi_s)_*\widehat{\eta}$. By item (1) above, η is Θ -invariant, and

$$(\text{proj}_{\Omega})_*\eta = (\text{proj}_{\Omega} \circ \pi_s)_*\widehat{\eta} = (\pi_b)_*(\text{proj}_{\Sigma_b})_*\widehat{\eta} = (\pi_b)_*\widehat{\mathbb{P}} = \mathbb{P}.$$

By Lemma 5.1(iii) π_s is N_η -to-one on a set of η -full measure. Moreover, recall that since $\mathbb{P}$ is fully supported on, π_b is $\mathbb{P}$ -a.s. one-to-one [14, p. 3.18]. Using that finite extensions preserve entropy, we have

$$\begin{aligned} h_\mu(\Theta | \mathbb{P}) &= h_\mu(\Theta) - h_\mathbb{P}(\theta) = h_{\widehat{\mu}}(\sigma_s) - h_{\widehat{\mathbb{P}}}(\sigma_b) = h_{\widehat{\mu}}(\sigma_s | \widehat{\mathbb{P}}) \\ &< h_{\widehat{\eta}}(\sigma_s | \widehat{\mathbb{P}}) = h_{\widehat{\eta}}(\sigma_s) - h_{\widehat{\mathbb{P}}}(\sigma_b) = h_\eta(\Theta) - h_\mathbb{P}(\theta) = h_\eta(\Theta | \mathbb{P}). \end{aligned}$$

Since $\eta(\mathcal{X}) = \widehat{\eta}(\pi_s^{-1}(\mathcal{X})) = \widehat{\eta}(\Sigma_s) = 1$, Proposition 2.9 implies that η is hyperbolic, which contradicts the maximality of μ for $\mathbb{P}$ -relative entropy. Therefore $\widehat{\mu}$ is a measure of $\widehat{\mathbb{P}}$ -maximal relative entropy. $\square$

For each $\mu \in \mathcal{M}_{\max}^{\mathcal{X}, \mathcal{X}}(\Theta | \mathbb{P})$ we have thus associated a measure $\widehat{\mu} \in \mathcal{M}_e(\sigma_s)$ such that $\widehat{\mu}$ is a measure of maximal $\widehat{\mathbb{P}}$ -relative entropy. Moreover, $(\pi_b \circ \text{proj}_{\Sigma_b})_*\widehat{\mu} = \mathbb{P}$. Define

$$\mathcal{M}_{\max}(\Sigma_s | \mathbb{P}) := \left\{ \widehat{\mu} \in \mathcal{M}_e(\sigma_s) ; \widehat{\mu} \text{ has maximal } \widehat{\mathbb{P}}\text{-relative entropy} \right\}$$

Then, the assignment

$$\mu \in \mathcal{M}_{\max}^{\mathcal{X}, \mathcal{X}}(\Theta | \mathbb{P}) \mapsto \widehat{\mu} \in \mathcal{M}_{\max}(\Sigma_s | \mathbb{P}) \quad (25)$$

is well defined and injective, since $(\pi_s)_*\widehat{\mu} = \mu$.

Hence, using the injectivity of (25), we obtain that

$$\# \mathcal{M}_{\max}^{\mathcal{X}, \mathcal{X}}(\Theta | \mathbb{P}) \leq \# \left(\left\{ \widehat{\mu} \in \mathcal{M}_e(\sigma_s) ; \widehat{\mu} \text{ has maximal } \widehat{\mathbb{P}}\text{-relative entropy} \right\} \right). \quad (26)$$

Observe that since $\mathbb{P}$ is fully supported on Ω , then $\widehat{\mathbb{P}}$ is fully supported on Σ_b (as a consequence of [14, Theorem 3.19 and Proposition 3.19]). From Corollary 4.5, with $X = \Sigma_s$, $Y = \Sigma_b = \text{proj}_{\Sigma_b}(\Sigma_s)$, $\pi = \text{proj}_{\Sigma_b}$ and $\nu = \widehat{\mathbb{P}}$ it follows that the set

$$\left\{ \widehat{\mu} \in \mathcal{M}_e(\sigma_s) : \widehat{\mu} \text{ has maximal } \widehat{\mathbb{P}}\text{-relative entropy} \right\}$$

is at most countable. Therefore from (26) we obtain that the set $\mathcal{M}_{\max}^{\mathcal{X}, \mathcal{X}}(\Theta | \mathbb{P})$ is countable, which completes the proof. $\square$

To prove Corollary B, we recall the following theorem for skew products, which refines Ruelle's inequality. We state a convenient weak version of the result presented in [28] that suffices for our purposes.

Theorem 5.3 ([28, Chapter 5, Theorem 3.1.1]). *Let*

$$\Theta : (\omega, x) \in \Omega \times M \mapsto (\theta(\omega), T_\omega(x)) \in \Omega \times M$$

be a $\mathcal{C}^{1+\alpha}$ skew-product diffeomorphism, and let $\mu \in \mathcal{M}_e(\Theta)$. Denote the base marginal by $\mathbb{P} := (\text{proj}_\Omega)_ \mu$. Let $\sigma(\mu) = \{\lambda_1, \dots, \lambda_n\}$ be the Lyapunov spectrum of Θ (see Definition 2.1). Define the set of fibre Lyapunov exponents*

$$\sigma(\mu | \mathbb{P}) := \left\{ \lambda \in \sigma(\mu); \text{proj}_{T_x M}(E_{(\omega, x)}^\lambda) \neq \{0\} \text{ for } \mu\text{-a.e. } (\omega, x) \in \Omega \times M \right\},$$

and for each $\lambda \in \sigma(\mu | \mathbb{P})$ set the (fibre) multiplicity

$$m_\lambda = \dim(\text{proj}_{T_x M} E_{(\omega, x)}^\lambda),$$

which is μ -a.s. constant by ergodicity.

Then

$$h_\mu(\Theta | \mathbb{P}) \leq \sum_{\substack{\lambda \in \sigma(\mu | \mathbb{P}) \\ \lambda > 0}} \lambda m_\lambda.$$

We can now prove Corollary B.

Proof of Corollary B. By virtue of Theorem A, it is sufficient to prove that every ergodic measure of $\mathbb{P}$ -relative maximal entropy of Θ is hyperbolic. In the notations of Theorem 5.3, it is clear that

$$\sigma(\mu) = \sigma(\theta) \cup \sigma(\mu | \mathbb{P}).$$

Since $\theta : \Omega \rightarrow \Omega$ is Axiom A then $0 \notin \sigma(\theta)$. Using that $\dim(M) = 2$ we have the cardinality of $\sigma(\mu | \mathbb{P})$ is either equal to 1 or 2. Therefore, to prove the corollary, it is enough to show that

$$\sigma(\mu | \mathbb{P}) = \{\lambda_-, \lambda_+\}$$

with $\lambda_- < 0 < \lambda_+$. From Theorem 5.3 we have that

$$0 < h_\mu(\Theta | \mathbb{P}) \leq \sum_{\substack{\lambda \in \sigma(\mu | \mathbb{P}) \\ \lambda > 0}} \lambda m_\lambda. \quad (27)$$

On the other hand, applying the same result to the skew product Θ^{-1} , we obtain that

$$0 < h_\mu(\Theta | \mathbb{P}) = h_\mu(\Theta^{-1} | \mathbb{P}) \leq \sum_{\substack{\tilde{\lambda} \in \sigma^{-1}(\mu | \mathbb{P}) \\ \tilde{\lambda} > 0}} \tilde{\lambda} m_{\tilde{\lambda}}, \quad (28)$$

where $\sigma^{-1}(\mu | \mathbb{P})$ denotes the set of the fibre Lyapunov exponents of Θ^{-1} . Since the fibrewise Lyapunov spectrum of Θ^{-1} is the sign-reversal of that of Θ , i.e. $\sigma^{-1}(\mu | \mathbb{P}) = \{-\lambda; \lambda \in \sigma(\mu | \mathbb{P})\}$. It follows from (28) that

$$0 < h_\mu(\Theta | \mathbb{P}) \leq \sum_{\substack{\tilde{\lambda} \in \sigma^{-1}(\mu | \mathbb{P}) \\ \tilde{\lambda} > 0}} \tilde{\lambda} m_{\tilde{\lambda}} = \sum_{\substack{\lambda \in \sigma(\mu | \mathbb{P}) \\ \lambda < 0}} -\lambda m_\lambda. \quad (29)$$

From (27) and (29) combined with $\dim(M) = 2$, we obtain that $\sigma(\mu | \mathbb{P}) = \{\lambda_-, \lambda_+\}$, which implies that μ is a hyperbolic measure and concludes the proof of the theorem. $\square$

We conclude this section by proving Proposition 1.6.

Proof of Proposition 1.6. Consider $G_k := \{x_1 \in \mathbb{S}^1; |2 - 2\pi k \sin(2\pi x_1)| \geq 9\}$. Choose k_0 large enough so that for every fixed $k \geq k_0$ we can pick disjoint intervals such that

$$\text{length}(J_1) = \text{length}(J_2) = \frac{1}{3}, \text{ and } J_1, J_2 \subset G_k.$$

Define the strips $S_1 := J_1 \times \mathbb{S}^1$ and $S_2 := J_2 \times \mathbb{S}^1 \subset \mathbb{T}^2$; see Figure 2. By construction, $\{2x_1 + x_2 + k \cos(2\pi x_1) + f(\omega); x_1 \in J_i\} = \mathbb{S}^1$, for $i \in \{1, 2\}$, every $x_2 \in \mathbb{S}^1$ and $\omega \in \mathbb{T}^2$, and the map $x_1 \in J_i \mapsto 2x_1 - x_2 + k \cos(2\pi x_1) + f(\omega)$ winds at least three times around $\mathbb{S}^1$ on each $i \in \{1, 2\}$, since $|2 - 2\pi k \sin(2\pi x_1)| \geq 9$ and $\text{length}(J_i) = 1/3$.

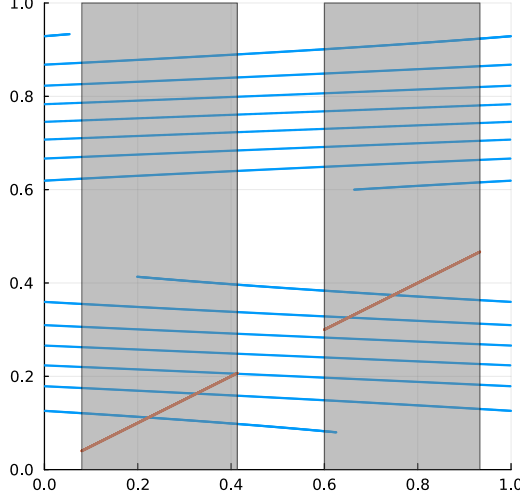


Figure 2. Illustration of the graph transform in Steps 1–2 for $k = 4$ and $\omega_0 = 0$. The grey strips indicate S_1 (left) and S_2 (right). The orange curves are the restrictions $\gamma_1 = \gamma|_{J_1} \in \mathcal{G}_1$ and $\gamma_2 = \gamma|_{J_2} \in \mathcal{G}_2$ of $\gamma(t) = (t, t/2)$, while the blue curves are their images $T_{\omega_0} \circ \gamma_1$ and $T_{\omega_0} \circ \gamma_2$.

Step 1. Define, for $i \in \{1, 2\}$, the family of graphs

$$\mathcal{G}_i := \{ \gamma : J_i \rightarrow \mathbb{T}^2 \text{ is } \mathcal{C}^1; \gamma(t) = (t, g(t)) \text{ with } \|g'\|_\infty < 1 \}.$$

Then, for k sufficiently large, for every $\omega \in \Omega$, every $\gamma \in \mathcal{G}_i$, and each $j \in \{1, 2\}$, there exists $\tilde{\gamma} = \tilde{\gamma}(\omega, \gamma, j) \in \mathcal{G}_j$ such that

$$\tilde{\gamma}(J_j) \subset T_\omega(\gamma(J_i)).$$

Indeed, for $\gamma(t) = (t, g(t))$ we compute

$$T_\omega \circ \gamma(t) = (2t - g(t) + k \cos(2\pi t) + f(\omega), t).$$

For $t \in J_i \subset G_k$,

$$\left| \frac{d}{dt} (2t - g(t) + k \cos(2\pi t) + f(\omega)) \right| \geq |2 - 2\pi k \sin(2\pi t)| - |g'(t)| \geq 9 - 1 = 8.$$

Hence $t \mapsto 2t - g(t) + k \cos(2\pi t) + f(\omega)$ is strictly monotone on J_i and, since $\text{length}(J_i) = 1/3$, its image wraps at least twice around the horizontal circle. Let $h : \mathbb{S}^1 \rightarrow J_i$ be any inverse branch of

$$t \in J_1 \mapsto 2t - g(t) + k \cos(2\pi t) + f(\omega) \in \mathbb{S}^1$$

Then

$$\{(s, h(s)); s \in J_j\} \subset \{T_\omega \circ \gamma(t); t \in J_i\}.$$

Define $\tilde{\gamma}(s) := (s, h(s))$. By the chain rule and the identity

$$h(2t - g(t) + k \cos(2\pi t) + f(\omega)) = t,$$

we obtain that

$$\sup_{y \in J_j} |h'(y)| = \sup_{t \in J_i} \left| \frac{1}{2 - g'(t) - 2\pi k \sin(2\pi t)} \right| \leq \frac{1}{8} < 1.$$

Therefore, $\tilde{\gamma} \in \mathcal{G}_j$ and Step 1 follows.

Step 2. Given a sequence $\underline{r} = (r_\ell)_{\ell \in \mathbb{Z}} \in \{1, 2\}^{\mathbb{Z}}$ and any $\omega \in \Omega$, there exists $x \in \mathbb{T}^2$ such that $T_\omega^\ell(x) \in S_{r_\ell}$, for every $\ell \in \mathbb{N}$.

Recall the definition of the family $\mathcal{G}_\ell$ from Step 1, with $\ell \in \{1, 2\}$. Consider the graph $\gamma_0: J_{r_0} \rightarrow \mathbb{T}^2$ defined by $\gamma_0(t) = (t, 0)$. Clearly $\gamma_0 \in \mathcal{G}_{r_0}$. By Step 1, there exists $\gamma_1: J_{r_1} \rightarrow \mathbb{T}^2$ with $\gamma_1 \in \mathcal{G}_{r_1}$ such that

$$\gamma_1(J_{r_1}) \subset T_\omega(\gamma_0(J_{r_0})).$$

Proceeding inductively, we construct graphs $\{\gamma_\ell: J_{r_\ell} \rightarrow \mathbb{T}^2\}_{\ell \in \mathbb{N}}$ with $\gamma_\ell \in \mathcal{G}_{r_\ell}$ such that for each $\ell \geq 1$,

$$\begin{aligned} \gamma_\ell(J_{r_\ell}) &\subset T_{\theta^{\ell-1}\omega} \circ \gamma_{\ell-1}(J_{r_{\ell-1}}) \\ &\subset \dots \subset T_{\theta^{\ell-1}\omega} \circ \dots \circ T_{\theta\omega} \circ T_\omega \circ \gamma_0(J_{r_0}) = T_\omega^\ell \circ \gamma_0(J_{r_0}). \end{aligned} \quad (30)$$

For each $\ell \in \mathbb{N}$ define the compact sets

$$\mathcal{T}_\ell := \{t \in J_{r_0}; T_\omega^j(\gamma_0(t)) \in S_{r_j} \text{ for every } j \in \{0, \dots, \ell\}\}.$$

By construction $\gamma_\ell \in \mathcal{G}_{r_\ell}$. Therefore, there exists $g_\ell: J_{r_\ell} \rightarrow \mathbb{R}$ such that $\gamma_\ell(t) = (t, g_\ell(t))$. Thus, $\gamma_\ell(J_{r_\ell}) = \{(t, g_\ell(t)); t \in J_{r_\ell}\} \subset S_{r_\ell}$, and from (30) we obtain that $(\mathcal{T}_\ell)_{\ell \in \mathbb{N}}$ is a nested sequence of nonempty compact sets, hence

$$\bigcap_{\ell=0}^{\infty} \mathcal{T}_\ell \neq \emptyset.$$

Choose $t_0 \in \bigcap_{\ell=0}^{\infty} \mathcal{T}_\ell$ and set $x := \gamma_0(t_0) = (t_0, 0)$. Then, for every $\ell \in \mathbb{N}$,

$$T_\omega^\ell(x) = T_\omega^\ell(\gamma_0(t_0)) \in S_{r_\ell},$$

which proves Step 2.

Step 3. We show that $h_{\text{top}}(\Theta \mid \mathbb{P}) > 0$, which completes the proof of the proposition.

Given $\omega \in \Omega$, define

$$X_\omega := \bigcap_{n=0}^{\infty} T_\omega^{-n}(J_1 \sqcup J_2).$$

By Step 2, $X_\omega \neq \emptyset$, and the coding map

$$\begin{aligned} h_\omega: X_\omega &\rightarrow \{1, 2\}^{\mathbb{N}} \\ x &\mapsto (J(T_\omega^i x))_{i \geq 0}, \end{aligned}$$

where $J(z) = i$ if $z \in J_i$ for $i \in \{1, 2\}$, is surjective.

Set $X_\Omega := \{(\omega, x); \omega \in \Omega \text{ and } x \in X_\omega\}$. Define the forward-time semi-conjugacy

$$\pi: (\omega, x) \in X_\Omega \mapsto (\omega, h_\omega(x)) \in \Omega \times \{1, 2\}^{\mathbb{N}}.$$

Let $\sigma: \{1, 2\}^{\mathbb{N}} \rightarrow \{1, 2\}^{\mathbb{N}}$ be the left shift and set

$$\sigma_\Omega: (\omega, s) \in \Omega \times \{1, 2\}^{\mathbb{N}} \rightarrow (A\omega, \sigma(s)) \in \Omega \times \{1, 2\}^{\mathbb{N}}.$$

By construction, $\pi \circ \Theta = \sigma_\Omega \circ \pi$. Since π is a surjective factor map and Θ and σ_Ω share the same base (Ω, A) , we obtain

$$h_{\text{top}}(\Theta \mid \mathbb{P}) \geq h_{\text{top}}(\sigma_\Omega \mid \mathbb{P}) = h_{\text{top}}(\sigma) = \log 2 > 0.$$

□

ACKNOWLEDGEMENTS

The authors acknowledge Yuri Lima for pointing out an inconsistency in the proof of Theorem 3.6 in an earlier version, and Anthony Quas for valuable discussions and for suggesting an improved approach for the proof of Theorem 4.4. GF and MMC are supported by an Australian Research Council Laureate Fellowship (FL230100088).

REFERENCES

- [1] M. Allahbakhshi, J. Antonioli, and J. Yoo. “Relative equilibrium states and class degree”. In: *Ergodic Theory Dynam. Systems* 39.4 (2019), pp. 865–888. DOI: [10.1017/etds.2017.50](https://doi.org/10.1017/etds.2017.50).
- [2] M. Allahbakhshi and A. Quas. “Class degree and relative maximal entropy”. In: *Trans. Amer. Math. Soc.* 365.3 (2013), pp. 1347–1368. DOI: [10.1090/S0002-9947-2012-05637-6](https://doi.org/10.1090/S0002-9947-2012-05637-6).
- [3] J. F. Alves, W. Bahsoun, M. Ruziboev, and P. Varandas. “Quenched decay of correlations for nonuniformly hyperbolic random maps with an ergodic driving system”. In: *Nonlinearity* 36.6 (2023), pp. 3294–3318. DOI: [10.1088/1361-6544/acd220](https://doi.org/10.1088/1361-6544/acd220).
- [4] E. Araujo, Y. Lima, and M. Poletti. “Symbolic dynamics for nonuniformly hyperbolic maps with singularities in high dimension”. In: *Mem. Amer. Math. Soc.* 301.1511 (2024), pp. vi+117. DOI: [10.1090/memo/1511](https://doi.org/10.1090/memo/1511).
- [5] A. Arbieto, C. Matheus, and K. Oliveira. “Equilibrium states for random non-uniformly expanding maps”. In: *Nonlinearity* 17.2 (2004), pp. 581–593. DOI: [10.1088/0951-7715/17/2/013](https://doi.org/10.1088/0951-7715/17/2/013).
- [6] J. Atnip, G. Froyland, C. González-Tokman, and S. Vaienti. “Equilibrium states for non-transitive random open and closed dynamical systems”. In: *Ergodic Theory and Dynamical Systems* 43.10 (2023), pp. 3193–3215. DOI: [10.1017/etds.2022.68](https://doi.org/10.1017/etds.2022.68).
- [7] J. Atnip, G. Froyland, C. González-Tokman, and S. Vaienti. “Thermodynamic Formalism and Perturbation Formulae for Quenched Random Open Dynamical Systems”. In: *Dissertationes Mathematicae* (2024). arXiv: [2307.00774](https://arxiv.org/abs/2307.00774) [math.DS].
- [8] J. Atnip, G. Froyland, C. González-Tokman, and S. Vaienti. “Thermodynamic Formalism for Random Weighted Covering Systems”. In: *Communications in Mathematical Physics* 386.2 (2021), pp. 819–902. DOI: [10.1007/s00220-021-04156-1](https://doi.org/10.1007/s00220-021-04156-1).
- [9] L. Barreira and Y. Pesin. *Nonuniform hyperbolicity*. Vol. 115. Encyclopedia of Mathematics and its Applications. Dynamics of systems with nonzero Lyapunov exponents. Cambridge University Press, Cambridge, 2007, pp. xiv+513. DOI: [10.1017/CB09781107326026](https://doi.org/10.1017/CB09781107326026).
- [10] S. Ben Ovadia. “Symbolic dynamics for non-uniformly hyperbolic diffeomorphisms of compact smooth manifolds”. In: *J. Mod. Dyn.* 13 (2018), pp. 43–113. DOI: [10.3934/jmd.2018013](https://doi.org/10.3934/jmd.2018013).
- [11] A. Blumenthal and L.-S. Young. “Equivalence of physical and SRB measures in random dynamical systems”. In: *Nonlinearity* 32.4 (2019), pp. 1494–1524. DOI: [10.1088/1361-6544/aafaa8](https://doi.org/10.1088/1361-6544/aafaa8).
- [12] T. Bogenschütz. “Entropy, pressure, and a variational principle for random dynamical systems”. In: *Random & Computational Dynamics* 1 (1992), pp. 99–116.
- [13] T. Bogenschütz and V. M. Gundlach. “Ruelle’s transfer operator for random subshifts of finite type”. In: *Ergodic Theory and Dynamical Systems* 15.3 (1995), pp. 413–447. DOI: [10.1017/S0143385700008464](https://doi.org/10.1017/S0143385700008464).
- [14] R. Bowen. *Equilibrium states and the ergodic theory of Anosov diffeomorphisms*. Ed. by J.-R. Chazottes. revised. Vol. 470. Lecture Notes in Mathematics. With a preface by David Ruelle. Springer-Verlag, Berlin, 2008, pp. viii+75.
- [15] R. Bowen. “Markov partitions for Axiom A diffeomorphisms”. In: *Amer. J. Math.* 92 (1970), pp. 725–747. DOI: [10.2307/2373370](https://doi.org/10.2307/2373370).
- [16] R. Bowen. “Some systems with unique equilibrium states”. In: *Math. Systems Theory* 8.3 (1974). Published 1974/75, pp. 193–202. DOI: [10.1007/BF01762666](https://doi.org/10.1007/BF01762666).
- [17] M. Brin and A. Katok. “On local entropy”. In: *Geometric Dynamics*. Ed. by J. Palis. Vol. 1007. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1983, pp. 30–38. DOI: [10.1007/BFb0061408](https://doi.org/10.1007/BFb0061408).

- [18] J. Buzzi. “Maximal entropy measures for piecewise affine surface homeomorphisms”. In: *Ergodic Theory Dynam. Systems* 29.6 (2009), pp. 1723–1763. DOI: [10.1017/S0143385708000953](https://doi.org/10.1017/S0143385708000953).
- [19] J. Buzzi, S. Crovisier, and Y. Lima. “Symbolic dynamics for large non-uniformly hyperbolic sets of three dimensional flows”. In: *Adv. Math.* 479 (2025), Paper No. 110410. DOI: [10.1016/j.aim.2025.110410](https://doi.org/10.1016/j.aim.2025.110410).
- [20] J. Buzzi, S. Crovisier, and O. Sarig. “Measures of maximal entropy for surface diffeomorphisms”. In: *Ann. of Math. (2)* 195.2 (2022), pp. 421–508. DOI: [10.4007/annals.2022.195.2.2](https://doi.org/10.4007/annals.2022.195.2.2).
- [21] J. Buzzi, S. Crovisier, and O. Sarig. *Strong positive recurrence and exponential mixing for diffeomorphisms*. 2025. arXiv: [2501.07455](https://arxiv.org/abs/2501.07455) [math.DS].
- [22] M. Denker, Y. Kifer, and M. Stadlbauer. “Thermodynamic formalism for random countable Markov shifts”. In: *Discrete and Continuous Dynamical Systems* 22.1-2 (2008), pp. 131–164. DOI: [10.3934/dcds.2008.22.131](https://doi.org/10.3934/dcds.2008.22.131).
- [23] T. Downarowicz. *Entropy in dynamical systems*. Vol. 18. New Mathematical Monographs. Cambridge University Press, Cambridge, 2011, pp. xii+391. DOI: [10.1017/CB09780511976155](https://doi.org/10.1017/CB09780511976155).
- [24] D. Dragičević, G. Froyland, C. González-Tokman, and S. Vaienti. “A spectral approach for quenched limit theorems for random hyperbolic dynamical systems”. In: *Transactions of the American Mathematical Society* 373.1 (2020), pp. 629–664. DOI: [10.1090/tran/7943](https://doi.org/10.1090/tran/7943).
- [25] V. M. Gundlach and Y. Kifer. “Expansiveness, specification, and equilibrium states for random bundle transformations”. In: *Discrete and Continuous Dynamical Systems* 6.1 (2000), pp. 89–120.
- [26] B. M. Gurevič. “Shift entropy and Markov measures in the space of paths of a countable graph”. In: *Dokl. Akad. Nauk SSSR* 192 (1970), pp. 963–965.
- [27] B. M. Gurevič. “Topological entropy of a countable Markov chain”. In: *Dokl. Akad. Nauk SSSR* 187 (1969), pp. 715–718.
- [28] B. Hasselblatt and A. Katok, eds. *Handbook of dynamical systems. Vol. 1B*. Elsevier B. V., Amsterdam, 2006, pp. xii+1222.
- [29] A. Katok. “Lyapunov exponents, entropy and periodic orbits for diffeomorphisms”. In: *Publications Mathématiques de l’IHÉS* 51 (1980), pp. 137–173. DOI: [10.1007/BF02684777](https://doi.org/10.1007/BF02684777).
- [30] A. Katok and B. Hasselblatt. *Introduction to the modern theory of dynamical systems*. Vol. 54. Encyclopedia of Mathematics and its Applications. With a supplementary chapter by Katok and Leonardo Mendoza. Cambridge University Press, Cambridge, 1995, pp. xviii+802. DOI: [10.1017/CB09780511809187](https://doi.org/10.1017/CB09780511809187).
- [31] Y. Kifer. “Equilibrium states for random expanding transformations”. In: *Random Comput. Dynam.* 1.1 (1992), pp. 1–31.
- [32] F. Ledrappier and L.-S. Young. “Entropy formula for random transformations”. In: *Probab. Theory Related Fields* 80.2 (1988), pp. 217–240. DOI: [10.1007/BF00356103](https://doi.org/10.1007/BF00356103).
- [33] F. Ledrappier and P. Walters. “A relativised variational principle for continuous transformations”. In: *Journal of the London Mathematical Society* s2-16.3 (1977), pp. 568–576. DOI: [10.1112/jlms/s2-16.3.568](https://doi.org/10.1112/jlms/s2-16.3.568).
- [34] Y. Lima. “Symbolic dynamics for one dimensional maps with nonuniform expansion”. In: *Ann. Inst. H. Poincaré C Anal. Non Linéaire* 37.3 (2020), pp. 727–755. DOI: [10.1016/j.anihpc.2019.10.001](https://doi.org/10.1016/j.anihpc.2019.10.001).
- [35] Y. Lima and C. Matheus. “Symbolic dynamics for non-uniformly hyperbolic surface maps with discontinuities”. In: *Ann. Sci. Éc. Norm. Supér. (4)* 51.1 (2018), pp. 1–38. DOI: [10.24033/asens.2350](https://doi.org/10.24033/asens.2350).
- [36] Y. Lima, J. C. Mongez, and J. P. Nascimento. *Symbolic dynamics for non-uniformly hyperbolic flows in high dimension*. 2025. arXiv: [2509.09050](https://arxiv.org/abs/2509.09050) [math.DS].
- [37] Y. Lima, D. Obata, and M. Poletti. *Measures of maximal entropy for non-uniformly hyperbolic maps*. 2024. arXiv: [2405.04676](https://arxiv.org/abs/2405.04676) [math.DS].
- [38] Y. Lima and O. M. Sarig. “Symbolic dynamics for three-dimensional flows with positive topological entropy”. In: *J. Eur. Math. Soc. (JEMS)* 21.1 (2019), pp. 199–256. DOI: [10.4171/JEMS/834](https://doi.org/10.4171/JEMS/834).
- [39] X. Liu. “Exponential decay of random correlations for random Anosov systems mixing on fibers”. In: *J. Differential Equations* 378 (2024), pp. 1–93. DOI: [10.1016/j.jde.2023.09.018](https://doi.org/10.1016/j.jde.2023.09.018).

- [40] A. Manoussos. “A Birkhoff type transitivity theorem for non-separable completely metrizable spaces with applications to linear dynamics”. In: *J. Operator Theory* 70.1 (2013), pp. 165–174. DOI: [10.7900/jot.2011may12.1971](https://doi.org/10.7900/jot.2011may12.1971).
- [41] K. Marin, M. Poletti, and F. Veiga. *Exponential mixing of measures of maximal entropy for certain skew products*. 2025. arXiv: [2509.09908](https://arxiv.org/abs/2509.09908) [math.DS].
- [42] V. Mayer, B. Skorulski, and M. Urbański. *Distance Expanding Random Mappings, Thermodynamical Formalism, Gibbs Measures and Fractal Geometry*. Vol. 2036. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 2011. DOI: [10.1007/978-3-642-23650-1](https://doi.org/10.1007/978-3-642-23650-1).
- [43] S. E. Newhouse. “Continuity properties of entropy”. In: *Ann. of Math. (2)* 129.2 (1989), pp. 215–235. DOI: [10.2307/1971492](https://doi.org/10.2307/1971492).
- [44] D. Obata. “Uniqueness of the measure of maximal entropy for the standard map”. In: *Comment. Math. Helv.* 96.1 (2021), pp. 79–111. DOI: [10.4171/cmh/508](https://doi.org/10.4171/cmh/508).
- [45] V. I. Oseledec. “A Multiplicative Ergodic Theorem: Lyapunov Characteristic Numbers for Dynamical Systems”. In: *Transactions of the Moscow Mathematical Society* 19 (1968), pp. 197–231.
- [46] W. Parry and M. Pollicott. “Zeta functions and the periodic orbit structure of hyperbolic dynamics”. In: *Astérisque* 187-188 (1990), p. 268.
- [47] J. B. Pesin. “Families of invariant manifolds that correspond to nonzero characteristic exponents”. In: *Izv. Akad. Nauk SSSR Ser. Mat.* 40.6 (1976), pp. 1332–1379, 1440.
- [48] K. Petersen, A. Quas, and S. Shin. “Measures of maximal relative entropy”. In: *Ergodic Theory Dynam. Systems* 23.1 (2003), pp. 207–223. DOI: [10.1017/S0143385702001153](https://doi.org/10.1017/S0143385702001153).
- [49] O. Sarig. “Lecture Notes on Thermodynamic Formalism for Topological Markov Shifts”. Lecture notes, Spring 2009. Version dated May 5, 2009. May 2009.
- [50] O. M. Sarig. “Symbolic dynamics for surface diffeomorphisms with positive entropy”. In: *J. Amer. Math. Soc.* 26.2 (2013), pp. 341–426. DOI: [10.1090/S0894-0347-2012-00758-9](https://doi.org/10.1090/S0894-0347-2012-00758-9).
- [51] O. M. Sarig. “Thermodynamic formalism for countable Markov shifts”. In: *Hyperbolic dynamics, fluctuations and large deviations*. Vol. 89. Proc. Sympos. Pure Math. Amer. Math. Soc., Providence, RI, 2015, pp. 81–117. DOI: [10.1090/pspum/089/01485](https://doi.org/10.1090/pspum/089/01485).
- [52] Y. G. Sinai. *Topics in Ergodic Theory*. Vol. 44. Princeton Mathematical Series. Princeton University Press, 1994.
- [53] M. Stadlbauer. “Coupling methods for random topological Markov chains”. In: *Ergodic Theory Dynam. Systems* 37.3 (2017), pp. 971–994. DOI: [10.1017/etds.2015.61](https://doi.org/10.1017/etds.2015.61).
- [54] M. Stadlbauer. “On random topological Markov chains with big images and preimages”. In: *Stoch. Dyn.* 10.1 (2010), pp. 77–95. DOI: [10.1142/S0219493710002863](https://doi.org/10.1142/S0219493710002863).
- [55] M. Stadlbauer, K. Suzuki, and P. Varandas. “Thermodynamic formalism for random non-uniformly expanding maps”. In: *Communications in Mathematical Physics* 385.1 (2021), pp. 369–427. DOI: [10.1007/s00220-021-04088-w](https://doi.org/10.1007/s00220-021-04088-w).
- [56] H. Takahasi. “Entropy-approachability for transitive Markov shifts over infinite alphabet”. In: *Proc. Amer. Math. Soc.* 148.9 (2020), pp. 3847–3857. DOI: [10.1090/proc/14637](https://doi.org/10.1090/proc/14637).
- [57] P. Walters. *Ergodic theory—introductory lectures*. Vol. 458. Lecture Notes in Mathematics. Springer-Verlag, Berlin-New York, 1975, pp. vi+198.
- [58] P. Walters. “Relative pressure, relative equilibrium states, compensation functions and many-to-one codes between subshifts”. In: *Trans. Amer. Math. Soc.* 296.1 (1986), pp. 1–31. DOI: [10.1090/S0002-9947-1986-0837796-8](https://doi.org/10.1090/S0002-9947-1986-0837796-8).
- [59] J. Yoo. “Decomposition of infinite-to-one factor codes and uniqueness of relative equilibrium states”. In: *J. Mod. Dyn.* 13 (2018), pp. 271–284. DOI: [10.3934/jmd.2018021](https://doi.org/10.3934/jmd.2018021).
- [60] J. Yoo. “Multiplicity structure of preimages of invariant measures under finite-to-one factor maps”. In: *Trans. Amer. Math. Soc.* 370.11 (2018), pp. 8111–8133. DOI: [10.1090/tran/7234](https://doi.org/10.1090/tran/7234).
- [61] Y. Zhu. “On local entropy of random transformations”. In: *Stoch. Dyn.* 8.2 (2008), pp. 197–207. DOI: [10.1142/S0219493708002275](https://doi.org/10.1142/S0219493708002275).